

Critical Care Nephrology

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CHAPTER OUTLINE

ACUTE KIDNEY INJURY IN THE CONTEXT OF CRITICAL ILLNESS, 1882
FLUID MANAGEMENT AND ITS IMPACT ON ACUTE KIDNEY INJURY, 1886

KIDNEY REPLACEMENT THERAPY, 1889
EXTRACORPOREAL BLOOD PURIFICATION, 1896

KEY POINTS

- Critical care nephrology is an evolving field that requires understanding of the pathobiology of critical illness as a multisystem condition characterized by crosstalk between multiple organs during injury, repair, and recovery.
- Critically ill in each patient involves the interplay of complex underlying comorbidities and specific acute challenges, leading to an individualized risk of incident acute kidney injury (AKI), AKI progression, requirement for kidney replacement therapy (KRT), and adverse post-AKI health care outcomes.
- There is no benefit to early goal-directed therapy for the management of sepsis, with regard to either mortality or prevention of AKI.
- Fluid therapy requires dynamic assessment of tolerance and responsiveness, incorporated into the dynamic timeline of phases of resuscitation, optimization, stabilization, and evacuation.
- The use of balanced crystalloids for resuscitation may reduce the risk of AKI in critical illness.
- In the context of AKI, accelerated (vs. standard) initiation of KRT is not recommended given the higher risk of complications including KRT dependence. Therefore the decision to initiate KRT should be individualized according to patient-specific solute and volume-control goals.
- The selection of KRT modality to support patients with AKI should consider available resources, cost, provider expertise, and patient needs.
- In the setting of acute liver failure or traumatic brain injury, continuous KRT (CRRT) is the preferred KRT modality to mitigate the risks of cerebral edema and herniation.
- The selection of CRRT modality to support critically ill patients with AKI should be considered according to available devices, protocol logistics, and provider expertise given that there is no objective evidence to favor one modality over the other.
- For critically ill patients with AKI receiving CRRT, key aspects of the CRRT prescription that should be standardized according to current evidence are 1. the prescription (on average) of a total effluent dose of 25 to 30 mL/kg/h; 2. the use of right internal jugular temporary catheters to minimize treatment interruptions; 3. the use of regional citrate anticoagulation—if feasible—to maximize filter life; and 4. the iterative monitoring of patient tolerance of fluid removal, fluid balance goals, and net ultrafiltration rate (UF_{NET}).
- Extracorporeal blood purification through adsorption-based technologies have gained prominence after the COVID-19 pandemic, although specific applications and protocols are the subject of ongoing investigation.

Studies now suggest that up to 60% of patients in intensive care units (ICUs) experience an episode of acute kidney injury (AKI).¹⁻³ In the most severe cases, kidney replacement therapy (KRT) is required. Short-term mortality among critically ill patients with AKI who require KRT is >50%⁴

and, among survivors, longevity and health-related quality of life are often poor.⁵ AKI and the resulting electrolyte and acid-base abnormalities occur on the background of multiple conditions that necessitate critical care. In this chapter, we first discuss AKI in the context of conditions

that are frequently encountered in the ICU including sepsis, cardiac surgery, liver failure, and acute respiratory distress syndrome (ARDS). This is followed by a discussion of fluid management and its relevance to the care of critically ill patients with AKI. We then review the principles of KRT prescription and delivery to critically ill patients with AKI. Finally, we discuss emerging technologies of blood purification for critically ill patients with AKI. We do not discuss acute brain injury in the ICU setting, which is associated with dysnatremias (including the syndrome of inappropriate secretion of antidiuretic hormone, cerebral salt wasting, and central diabetes insipidus). The reader is referred to [Chapter 14](#) for further discussion of these disorders.

ACUTE KIDNEY INJURY IN THE CONTEXT OF CRITICAL ILLNESS

A list of common etiologies and risk factors for AKI, specifically in ICU settings, is provided in [Table 64.1](#). Selected clinical contexts of AKI in critically ill adult patients are discussed as follows.

SEPSIS-ASSOCIATED ACUTE KIDNEY INJURY

Sepsis is the most common cause of AKI in the ICU, contributing to approximately 50% of cases.^{2,4,6} A large epidemiologic study found that sepsis-associated AKI (SA-AKI) was present in one of six ICU patients.⁷ SA-AKI is associated with substantially higher mortality than

sepsis without AKI or nonseptic AKI.⁶⁻⁹ Notably, the relationship between AKI and sepsis appears bidirectional, as accumulating data suggest that AKI is a risk factor for the subsequent development of secondary infections and de novo sepsis.¹⁰⁻¹² In 2023 a definition for SA-AKI was proposed.¹³ It combines the consensus criteria for both sepsis, using the sepsis-3 definition¹⁴ and the Kidney Disease: Improving Global Outcomes (KDIGO) AKI criteria.¹⁵ The traditional understanding of the pathophysiology of SA-AKI was that sepsis triggered acute tubular injury via global hypoperfusion.¹⁶ However, both animal and human studies have demonstrated that, while renal blood flow (RBF) appears to decline later in established SA-AKI, macrovascular RBF is increased in early SA-AKI, with relatively preserved histology with minimal or patchy tubular injury without necrosis.¹⁷⁻²⁵ The pathophysiology of SA-AKI is complex and heterogeneous, involving microcirculatory dysfunction, inflammation, oxidative stress, mitochondrial dysfunction, altered cellular bioenergetics, and endothelial dysfunction.²⁶⁻²⁸ In this context, one should recognize that while SA-AKI often coexists with ischemic acute tubular necrosis (ATN) in the setting of overt shock, sepsis without hypotension can also cause AKI.

Despite multiple prior and ongoing studies (NCT00711789, NCT02664753, NCT03901807, NCT05900284),²⁹ therapeutic trials specifically targeting SA-AKI have thus far failed to demonstrate clear benefits.³⁰⁻³² Thus management of SA-AKI remains largely supportive, with timely antibiotics, infectious source control, and hemodynamic optimization.

ACUTE KIDNEY INJURY IN THE SETTING OF HEART DISEASE

Since cardiorenal syndromes (CRS) are discussed in detail in [Chapter 42](#), we focus on AKI in the setting of cardiac surgery or mechanical cardiopulmonary support.

CARDIAC SURGERY-ASSOCIATED ACUTE KIDNEY INJURY

Cardiac surgery-associated AKI (CSA-AKI) is the second most common cause of AKI in the ICU and is associated with increased mortality.^{4,33-36} CSA-AKI is defined as AKI occurring 2 to 7 days after surgery and is reported, by KDIGO criteria, to complicate 20% to 30% of cardiac surgeries.³⁶⁻³⁸ The pathophysiology of CSA-AKI is multifactorial including hypoperfusion, inflammation, oxidative stress, atheroembolism, hemolysis, nephrotoxin exposure, and exposure to cardiopulmonary bypass (CPB).^{36,37} The duration of CPB is a significant predictor of AKI, but extensive randomized controlled trials (RCTs) aimed at avoiding or reducing CPB have yielded mixed results.³⁹⁻⁴¹ Similarly, RCTs of therapeutics to prevent or treat CSA-AKI have failed to result in benefits.⁴²⁻⁴⁵ Single-center RCTs^{46,47} have found that perioperative nitric oxide (NO) delivered via the CPB circuit or ventilator decreases the rate of CSA-AKI or chronic kidney disease (CKD), presumably via correction of hemolysis-induced depletion of renal NO. Though promising, the use of NO to prevent CSA-AKI remains investigational and additional RCTs are ongoing (NCT04216927, NCT05757557).^{37,48} The use of biomarkers to prevent CSA-AKI also holds promise. The multicenter PrevAKI study randomized 278 postoperative patients at

Table 64.1 Common Etiologies and Risk Factors for Acute Kidney Injury (AKI) in the Intensive Care Unit

Etiologies/Contributing Factors ^{2,4a}	Risk Factors/Predisposing Conditions ^{2,515}
• Sepsis/septic shock: 40%-50% • Major surgery: 35% • Volume depletion: 25%-35% • Cardiogenic shock: 15%-25% • Drug toxicity: 15%-20% • Hepatorenal syndrome: 3%-6% • Urinary tract obstruction: 1%-3%	• Chronic kidney disease • Prior AKI • Advanced age • Frailty⁵¹⁶ • Nonrenal comorbidities: • Diabetes • Heart disease, especially CHF • Liver disease • Malignancy • Treatment with diuretics, NSAIDs, or RAAS inhibitors

^aThe frequencies of these etiologies/factors are derived from Uchino S, Kellum JA, Bellomo R, et al. Acute renal failure in critically ill patients: a multinational, multicenter study. *JAMA*. 2005;294(7):813–818 and Hoste EA, Bagshaw SM, Bellomo R, et al. Epidemiology of acute kidney injury in critically ill patients: the multinational AKI-EPI study. *Intensive Care Med*. 2015;41(8):1411–1423 but are rounded for simplicity.

CHF, Congestive heart failure; NSAID, nonsteroidal anti-inflammatory drug; RAAS, renin-angiotensin-aldosterone system.

high risk of CSA-AKI, identified on the basis of elevated urinary tissue inhibitor of metalloproteinases 2 (TIMP-2) * insulin-like growth factor binding protein 7 (IGFBP7), to standard care or the implementation of a KDIGO-based AKI bundle, which includes minimization of nephrotoxic medications, prevention of hyperglycemia, and volume optimization via hemodynamic monitoring.⁴⁹ Though the need for KRT and mortality were similar in both groups, the rate of stage 2-3 AKI was significantly reduced.⁴⁹

ACUTE KIDNEY INJURY IN PATIENTS REQUIRING MECHANICAL CARDIOPULMONARY SUPPORT

The use of extracorporeal membrane oxygenation (ECMO) in adults grew exponentially after the 2009 H1N1 influenza outbreak, with overall ECMO usage in adults more than tripling between 2008 and 2014.⁵⁰ Notably, 2023 guidelines recommend the use of venovenous (VV-) ECMO for refractory ARDS.⁵¹

Up to 85% of adults on ECMO develop AKI, with roughly 45% requiring KRT.⁵²⁻⁵⁷ As in other settings, AKI in patients requiring ECMO is independently associated with mortality.⁵²⁻⁵⁶ AKI develops more commonly in patients receiving venoarterial- (VA-) ECMO for cardiac failure than those on VV-ECMO for isolated respiratory failure.⁵⁶⁻⁵⁸ Observational data suggest that timely KRT may be beneficial in preventing or mitigating volume overload in ECMO patients.^{59,60} Fluid overload in ECMO patients is independently associated with increased mortality⁶¹⁻⁶⁴, and guidelines⁶⁵ suggest targeting euolemia with diuretics or KRT once feasible to facilitate ECMO weaning. However, experts recommend against *preemptive* KRT use in patients on ECMO, given a lack of data for benefit of KRT beyond the usual indications.⁶⁶

The pathogenesis of AKI on ECMO is complex. ECMO-specific factors that contribute to AKI include inflammation and disordered coagulation induced by the circuit, pigment injury from hemolysis, and nonpulsatile blood flow in VA-ECMO.⁶⁷ Excess circuit-related hemolysis may manifest with pink urine or KRT effluent and can be confirmed by measuring plasma-free hemoglobin levels.^{61,68-70}

Like ECMO, the use of left ventricular assist devices (LVADs) to treat end-stage heart failure has grown dramatically during the past 2 decades.^{71,72} On average, LVAD placement produces an initial improvement in kidney function, likely reflecting improved renal perfusion in patients with type 2 CRS (i.e., kidney dysfunction due to chronic cardiac dysfunction), followed by a gradual decline back toward baseline.⁷³⁻⁷⁶ However, kidney function may not improve in those with intrinsic kidney disease. Like patients receiving ECMO, patients managed with LVAD are at elevated risk for AKI throughout their course, and AKI (particularly AKI requiring KRT) is associated with poor outcomes.⁷⁵⁻⁷⁸ LVAD-specific causes of AKI include hemolysis, right ventricular failure, and possibly nonpulsatile blood flow.^{75,76}

ACUTE KIDNEY INJURY IN THE SETTING OF RESPIRATORY FAILURE AND ACUTE RESPIRATORY DISTRESS SYNDROME

Acute kidney injury is an independent risk factor for respiratory failure and vice versa, and the presence of

either condition complicating the other is associated with a greater than twofold increased risk of death.⁷⁹⁻⁸² Though traditionally attributed to fluid overload, the high mortality of combined respiratory and kidney failure may be mediated by kidney-lung crosstalk. This crosstalk involves a variety of complex pathophysiologic mechanisms including the inflammatory effects of AKI on the lung endothelium mediated by multiple cellular and cytokine mediators (e.g., interleukin-6), impaired alveolar fluid clearance due to downregulation of pulmonary sodium and water channels, and the deleterious hemodynamic effects of mechanical ventilation on RBF and microvascular flow.^{66,82,83}

ARDS is among the most severe forms of respiratory failure with an in-hospital mortality rate of approximately 40%.⁸⁴ ARDS is defined as acute-onset pulmonary edema not explained by cardiac failure or fluid overload resulting in hypoxemia with a PaO₂-to-FiO₂ ratio of ≤ 300 mm Hg.^{85,86} Treatment consists of addressing the underlying cause and providing lung protective mechanical ventilation (lung protective ventilation) with limitations in tidal volume and inspiratory pressures and tolerance of hypercapnia, a ventilatory strategy that was shown in the ARDS Network trial to reduce mortality.⁸⁷ Patients in the lung protective ventilation group had less renal dysfunction,⁸⁷ consistent with preclinical and epidemiologic data that injurious mechanical ventilation may induce AKI.⁸⁸⁻⁹⁰ Permissive hypercapnia normally triggers compensatory renal bicarbonate reabsorption.

Achieving permissive hypercapnia without causing severe acidemia in patients with AKI may necessitate supplemental bicarbonate or KRT,^{87,91} especially in severe ARDS when respiratory carbon dioxide (CO₂) elimination is severely impaired. However, the volume load of sodium bicarbonate may be prohibitive in patients with severe hypoxemia,⁹¹ and rapid administration of bicarbonate may abruptly worsen hypercarbia.⁹² If volume overload is present, correction of acidemia may be preferred through KRT. An alternative method to correct acidosis with lung protective ventilation is using an extracorporeal CO₂ removal (ECCO₂R) device, although trial data in ARDS are lacking.⁹³

Avoidance of fluid overload in ARDS is essential. A conservative fluid management strategy (i.e., less fluid/more diuretics) in ARDS was shown in a large multicenter RCT to result in shorter duration of mechanical ventilation and ICU stay without affecting mortality.⁹⁴ The conservative arm also had a trend toward decreased need for KRT (10 vs. 14%, $P = 0.06$)⁹⁴ and, when adjusted post hoc for the effect of fluid balance on serum creatinine, a decreased rate of AKI.⁹⁵ KRT in patients with ARDS without AKI to achieve negative or even fluid balance has been studied in some small trials and, though controversial, may be considered for refractory hypoxemia in some instances.⁹⁶⁻⁹⁸

ACUTE KIDNEY INJURY IN THE SETTING OF LIVER DISEASE

ACUTE KIDNEY INJURY IN THE SETTING OF CIRRHOSIS AND HEPATORENAL SYNDROME

Hepatorenal syndrome (HRS), particularly HRS-AKI, which was previously referred to as type 1 HRS, is a complication of decompensated cirrhosis associated with high mortality in which the vascular and neurohormonal disturbances of

advanced liver disease and portal hypertension result in profound renal hypoperfusion unresponsive to intravenous (IV) albumin or other fluids.⁹⁹⁻¹⁰² In modern cohorts, HRS-AKI accounts for 12% of AKI cases in the setting of cirrhosis, whereas prerenal AKI (44%) and ATN (30%) are significantly more common.¹⁰³ Notably, the mortality of HRS and ATN in the setting of cirrhosis is similar at approximately 50% at 90 days.^{103,104}

Developments in the diagnosis of HRS-AKI include adoption of the KDIGO criteria for AKI diagnosis.^{99,105} In addition, though these urinary indices tend to perform poorly in ICU patients in differentiating prerenal AKI from ATN,¹⁰⁶ cutoffs of fractional excretion of sodium (FENa) and urea (FeUrea) have been suggested to differentiate HRS from ATN or prerenal AKI (with FENa <0.2% and FeUrea <28.2% being consistent with HRS-AKI and <0.1% and <21%, respectively, being especially suggestive).¹⁰⁷⁻¹⁰⁹ Furthermore, though the absence of proteinuria >500 mg/g creatinine remains a traditional diagnostic criterion for HRS, urinary albumin has been shown in studies to be useful in differentiating ATN and HRS, with albuminuria >100 mg/g creatinine being suggestive of ATN.^{109,110} Likewise, studies suggest that the tubular injury biomarker neutrophil gelatinase-associated lipocalin (NGAL) may help differentiate HRS and ATN, with a urinary NGAL level >220 to 250 µg/g creatinine being consistent with ATN.¹⁰⁹⁻¹¹² Despite these diagnostic advances, the diagnosis of HRS-AKI in clinical practice remains a tremendous challenge. Though the diagnostic criteria imply that HRS is a diagnosis of exclusion, HRS can coexist with ATN, CRS from cirrhotic cardiomyopathy, abdominal compartment syndrome from tense ascites, or other unrelated causes of AKI or CKD.¹⁰⁰⁻¹⁰²

Terlipressin has been validated as an effective treatment for HRS-AKI in the Study to Confirm Efficacy and Safety of Terlipressin in Hepatorenal Syndrome (HRS) Type 1 (CONFIRM).¹¹³ Though CONFIRM demonstrated a significant increase in HRS reversal (29% vs. 16% in placebo arm), the terlipressin arm also experienced an increased risk of adverse effects including respiratory failure (10% with terlipressin vs. 3% in the placebo arm) and death due to respiratory failure (11% vs. 2%, respectively). This is thought to result from excessive albumin use and the regional hemodynamic effects of terlipressin including intense systemic vasoconstriction, pulmonary venoconstriction, lack of pulmonary arterial vasoconstriction, and absent inotropic effect.¹¹⁴ The positive primary outcome of the CONFIRM trial resulted in FDA approval of terlipressin in 2022 with a black box warning about the risk of fatal respiratory failure. Contraindications to terlipressin include any hypoxemia or need for supplemental oxygen, acute-on-chronic liver failure (ACLF) grade 3, and serum creatinine >5.0 mg/dL, as these patients are at increased risk of respiratory failure or treatment nonresponse.^{108,113,115,116} Norepinephrine is recommended as an alternative to terlipressin, as it appears to have similar efficacy in achieving HRS reversal, but it requires ICU admission in most centers.^{99,108} Octreotide and midodrine are not recommended because they are rarely effective and may cause harm by delaying effective therapy.^{99,108,113}

The results of CONFIRM and a second large RCT published in 2021 examining albumin use in hospitalized patients with cirrhosis¹¹⁷ highlight the risk of harmful

volume overload from excessive albumin administration in patients with liver disease. Similarly, the association between intraabdominal hypertension (IAH) and AKI in cirrhosis, as well as the potential for improved kidney function through cautious therapeutic paracentesis, is gaining recognition.¹¹⁸⁻¹²¹ Therefore while albumin remains the initial treatment for most cirrhosis patients with AKI, it is important to first assess their volume status thoughtfully. Furthermore, given data that delay in implementation of effective vasoconstrictor therapy increases the risk of treatment failure,^{99,108,113} those who felt likely to have HRS may benefit from initiation of vasoconstrictors before failing 48 hours of albumin therapy.^{100-102,122}

ACUTE LIVER FAILURE

Acute liver failure (ALF), previously known as fulminant hepatic failure, is defined by 1. hepatic encephalopathy, 2. coagulopathy (international normalized ratio [INR] ≥1.5), 3. onset of illness <26 weeks, and 4. no evidence of cirrhosis.^{123,124} The distinction between ALF and chronic liver disease is vitally important because some of the complications and management options apply specifically to ALF. Common causes of ALF include toxic, viral, and ischemic liver injury, with acetaminophen toxicity alone accounting for about 50% in modern cohorts.¹²³⁻¹²⁵ Though some cases will spontaneously resolve, the mortality of ALF is high with supportive care alone.¹²³ Complications of ALF overlap with decompensated cirrhosis and include hepatic encephalopathy AKI, coagulopathy and bleeding, hypoglycemia, infection and sepsis, and multiorgan failure.¹²⁵ However, in contrast to cirrhosis, in which clinically significant cerebral edema is characteristically absent,^{126,127} severe hepatic encephalopathy from ALF is frequently accompanied by cerebral edema with progressive intracranial hypertension and risk of lethal brainstem herniation. Cerebral edema is present in up to 35% of patients with ALF and grade 3 hepatic encephalopathy and 75% of patients with grade 4 hepatic encephalopathy.¹²³ First-line treatment for confirmed or suspected intracranial hypertension is hyperosmolar therapy with mannitol, though it requires monitoring of serum osmolality and its use may be limited by renal dysfunction in those not on KRT.^{123,128} Similarly, hypertonic saline to achieve serum sodium of 145 to 155 mEq/L is recommended in patients with or at high risk of cerebral edema.^{123,129}

AKI in the setting of ALF may be due to renal hypoperfusion (e.g., sepsis, bleeding, and HRS) or intrinsic injury (e.g., ischemic ATN, direct tubular toxicity of acetaminophen or *Amanita*) and is associated with increased risk of cerebral edema and overall poor prognosis.¹²³ For a variety of reasons, patients with ALF may require KRT relatively early in the course of their AKI.¹²⁵ Typically, continuous renal replacement therapy (CRRT) is preferred to intermittent hemodialysis (IHD) to minimize the risk of hypotension and increase in intracranial pressure (ICP), which can combine to severely reduce cerebral perfusion pressure. CRRT also offers more flexibility to manage volume status, clear ammonia, and precisely manage hyponatremia, which frequently accompanies ALF and may aggravate cerebral edema.^{123,130-132} In contrast to chronic liver disease, KRT does appear to have a role in specifically treating hyperammonemia in ALF in the absence of AKI,

particularly when ammonia is >250 $\mu\text{g/L}$, with higher-than-standard doses of CRRT typically required to effectively clear ammonia.^{131,133} Beyond the risk of hypotension and increased ICP, a preference for CRRT over IHD in ALF is supported by limited observational and trial data.^{124,130}

ABDOMINAL COMPARTMENT SYNDROME

Intraabdominal hypertension and abdominal compartment syndrome are increasingly recognized as common causes of decreased kidney function in ICU patients.¹³⁴⁻¹³⁷ Intraabdominal hypertension is defined as a sustained elevation in intraabdominal pressure (IAP) ≥ 12 mm Hg, and abdominal compartment syndrome is defined as sustained IAP >20 mm Hg associated with new organ dysfunction. Intraabdominal hypertension has been reported in 30% to 50% of mixed medical-surgical ICU patients, with abdominal compartment syndrome occurring in 5% to 12%.¹³⁵⁻¹³⁸ Risk factors for IAH include trauma, major burns, abdominal surgery, mechanical ventilation, obesity, ascites, hemoperitoneum, gastric or bowel distention, large-volume resuscitation, and pancreatitis.¹³⁸ Physical examination is unreliable for diagnosis.^{134,138,139} Instead, to estimate IAP, 25 to 30 mL of sterile water is instilled into the urinary bladder via a Foley catheter and pressure tubing is used to transduce bladder pressure at end-expiration with the patient relaxed in the supine position.¹³⁹

IAH can decrease the perfusion of any abdominal organ, and effects can be transmitted to other compartments leading to decreased cardiac output (CO), impaired ventilation, and increased ICP. The kidneys are particularly sensitive such that abdominal compartment syndrome is thought to be unlikely in the absence of oliguria. Decreased kidney function from IAH appears to be due to reduced perfusion mediated primarily by elevations in renal venous and parenchymal pressures (rather than by decreased CO or ureteral compression).^{138,139} Ultimately, a functional prerenal state develops with neurohormonal activation, renal vasoconstriction, renal sodium retention, and oliguria. IAH has been implicated in the pathogenesis of both CRS and HRS, with small studies in both settings suggesting that paracentesis may improve renal function.^{118-121,139,140}

In established abdominal compartment syndrome, decompressive laparotomy remains the treatment of choice despite its high morbidity.^{134,138,139} Less invasive measures to control IAH include sedation and potentially neuromuscular blockade to decrease abdominal muscle tone and decompression through paracentesis or nasogastric suction. Fluid management is complex in the setting of IAH. While decreased CO in abdominal compartment syndrome can worsen with hypovolemia, excessive fluid intake is associated with a higher risk of IAH. Any benefit from fluids following initial resuscitation is likely to be temporary, and giving too much fluid may trigger a harmful cycle where fluids are administered for a short-term improvement in CO but lead to increased IAH and subsequent reduced CO, prompting further fluid administration.¹³⁷ Animal model data suggest that IV fluids may restore CO but do not restore organ function in abdominal compartment syndrome.¹⁴¹ Traditional measures of volume status can be unreliable in abdominal compartment syndrome, as central venous and pulmonary artery occlusion pressures are increased, and

ultrasound may show a collapsed inferior vena cava, all due to IAH. Small case series suggest that CRRT can lower IAP, but the role of KRT for patients with IAH without kidney dysfunction remains unclear.^{139,142,143} When providing KRT to patients with IAH, the femoral vein should be avoided for dialysis catheter placement because IAH may lead to recirculation and impaired clearance.¹⁴⁴

KEY CONCEPTS IN NEPHROTOXIC ACUTE KIDNEY INJURY

In critically ill patients with AKI, medications must be reviewed frequently in order to 1. prevent worsening or prolongation of AKI due to avoidable nephrotoxic exposure, 2. prevent toxic accumulation of nonnephrotoxic drugs that may cause extrarenal toxicity in the presence of impaired kidney function (e.g., neurotoxicity from morphine, gabapentinoids, or cefepime),¹⁴⁵⁻¹⁴⁷ and 3. ensure proper drug dosing for patients receiving KRT. The topics of drug-induced acute interstitial nephritis (AIN), drug dosing in AKI, and kidney disease from anticancer drugs are reviewed in detail elsewhere (Chapters 37, 56, and 57, respectively). We focus on two evolving topics in drug-induced acute tubular injury in critically ill patients.

VANCOMYCIN-ASSOCIATED ACUTE KIDNEY INJURY

Though controversy persists given inherent limitations in observational data, vancomycin has been increasingly recognized as a contributor to AKI in critically ill patients.¹⁴⁸⁻¹⁵⁰ The nephrotoxicity of vancomycin was initially attributed to the impurities of early formulations, referred to as “Mississippi mud.”¹⁴⁸ Though current purified formulations are clearly less toxic than prior preparations, increasing rates of methicillin-resistant *Staphylococcus aureus* (MRSA) infections¹⁵¹ and the concomitant increase in vancomycin use have led to increasingly frequent reports of vancomycin-associated AKI (VA-AKI) with modern formulations.¹⁴⁸ In meta-analyses, the reported rates of VA-AKI ranged from 5% to 43% of vancomycin-treated patients with a relative risk of AKI with vancomycin of 2.45 compared with other antimicrobials (mostly linezolid).^{152,153} VA-AKI has a mean onset of 4 to 17 days after drug initiation and approximately 75% of cases improve or resolve by hospital discharge.^{148,152}

Vancomycin undergoes no metabolism and is entirely cleared unchanged by the kidneys.¹⁴⁸ The mechanism of VA-AKI appears to be toxic acute tubular injury caused by oxidative injury and obstructive tubular casts.^{148,154,155} The casts are composed of noncrystal nanospheric vancomycin aggregates embedded in uromodulin, as demonstrated by immunohistochemistry of human kidney biopsy and urinary sediment specimens, and reproduced in mouse models.^{154,155} Risk factors include both patient factors—baseline kidney function, obesity, severity of illness, concomitant nephrotoxin exposure (e.g., aminoglycosides), and age—and drug factors like higher target trough levels (i.e., >15 mg/L, as used for severe MRSA infections); higher total daily dose (i.e., >4 g per day); intermittent dosing (rather than continuous infusions); and longer duration of therapy (i.e., >7 days).^{148,152,153,156-159}

An additional reported risk factor for VA-AKI is concomitant exposure to piperacillin-tazobactam (P-TZ),

which alone is not generally considered nephrotoxic (except for rare cases of AIN). Though not found in all studies, a signal for increased rates of AKI with this combination compared with vancomycin alone has been repeatedly detected in the literature.¹⁶⁰⁻¹⁶⁸ However, with some exceptions,¹⁶⁰ the majority of the studies tying the combination of P-TZ and vancomycin to AKI have failed to detect a difference in need for KRT or mortality,¹⁶⁷⁻¹⁷² outcomes strongly linked to AKI. Moreover, the mechanism of this vancomycin-specific additive nephrotoxicity of P-TZ is unknown and multiple attempts to replicate this synergistic nephrotoxicity in animal models and in vitro studies have failed.¹⁷³⁻¹⁷⁵ One proposed explanation for these contradictory findings is that P-TZ, which is excreted via organic anion transporters,¹⁷⁶ inhibits tubular secretion of creatinine, leading to increased rates of creatinine-defined pseudo-AKI without any associated tubulopathy or morbidity. Support for this theory comes from a 2022 biomarker study of nearly 200 ICU patients treated with vancomycin and P-TZ or vancomycin and cefepime. The P-TZ-vancomycin combination was associated with higher rates of creatinine-defined AKI, but there was no association with AKI as defined by cystatin C, need for KRT, or mortality.¹⁷⁷ Subsequently, the 2023 Antibiotic Choice on Renal Outcomes (ACORN) study randomized more than 2500 adults hospitalized for acute infection to cefepime or P-TZ and, though 77% of the participants were also on vancomycin at enrollment, the rates of AKI or death were similar in both arms while the cefepime arm had increased rates of delirium or coma.¹⁷⁸ Considering that kidney dysfunction is a major risk factor for cefepime neurotoxicity,¹⁴⁷ these results suggest that cefepime should not be used preferentially to PT-Z in combination with vancomycin or in kidney dysfunction. However, as observational data suggest that synergistic nephrotoxicity is more likely with longer courses,^{160,179} caution may be warranted with prolonged courses of PT-Z and vancomycin.

ACUTE KIDNEY INJURY ASSOCIATED WITH INTRAVENOUS IODINATED CONTRAST

Emerging data suggest that IV iodinated contrast, as given with computed tomography (CT), is less nephrotoxic than previously thought. Although a few exceptions exist,^{180,181} more than a dozen studies and meta-analyses¹⁸²⁻¹⁹⁶ analyzing thousands of patients across a variety of settings have been published since 2013, suggesting that modern low- or iso-osmolar IV contrast exposure has little to no nephrotoxicity. Many of these analyses focused on patients at higher risk of contrast-associated AKI (CA-AKI) due to underlying CKD. A 2023 multicenter analysis of >14,000 patient encounters, using weighting by propensity scores and entropy balancing to mitigate confounding and biases, found no increased risk of persistent kidney dysfunction or need for KRT in patients with established AKI before IV contrast exposure.¹⁹¹ While the risk of CA-AKI and relevant AKI outcomes after IV contrast administration is far lower than previously thought, the risk may not be zero, and measures to reduce the risk, such as minimization of contrast exposure and provision of IV fluids when appropriate, should still be pursued.¹⁹⁷⁻²⁰⁰

Importantly, nearly all of these studies questioning the nephrotoxicity of iodinated contrast analyzed IV rather than

intraarterial contrast exposure. Both modern and historical data support the notion that intraarterial contrast exposure is more nephrotoxic, though many have speculated that this differential risk relates primarily to procedural aspects of angiography, such as the risk of subclinical atheroembolism, rather than enhanced toxicity related to the route of contrast administration.¹⁹⁹⁻²⁰⁵ Regardless of the mechanism, the risk of AKI after intraarterial contrast exposure needs to be carefully considered in at-risk patients, and implementation science to mitigate the risk of AKI after cardiac catheterization has shown promise.^{206,207}

Observational studies examining coronary angiography in advanced CKD patients with acute coronary syndromes reveal that although CKD patients face a higher risk of AKI or death compared to non-CKD patients, those with CKD who undergo angiography have lower mortality rates than CKD patients who do not receive indicated angiography.²⁰⁸⁻²¹¹ Though these observational studies are subject to residual confounding and biases, they suggest that withholding indicated diagnostic tests or interventions in patients with renal dysfunction due to excessive concern about AKI may inadvertently lead to worse outcomes, a concept termed “renalism.” Though data demonstrating that “renalism” exists in ICU populations are lacking, many experts^{199,200,203,205,212} and 2020 consensus guidelines²¹³ recommend performing studies using IV contrast regardless of kidney function when needed for definitive diagnosis or treatment of life-threatening conditions (e.g., sepsis or suspected pulmonary embolism).

FLUID MANAGEMENT AND ITS IMPACT ON ACUTE KIDNEY INJURY

VOLUME MANAGEMENT IN THE INTENSIVE CARE UNIT

The approach to fluid administration in the ICU has undergone an evolution from the adage that patients “must swell to get well”—driven by the influence of the landmark 2001 single-center early goal-directed therapy (EGDT) trial by Rivers et al.²¹⁴—to an increasing appreciation that volume overload is a marker and potentially a mediator of poor outcomes in critical illness. Three subsequent large multicenter RCTs, ProCESS (Protocol-Based Care for Early Septic Shock), ARISE (Australasian Resuscitation in Sepsis Evaluation), and ProMiSe (Protocolised Management in Sepsis), collectively randomized more than 4000 patients with septic shock and demonstrated no benefit of EGDT over standard care.²¹⁵⁻²¹⁷ Furthermore, observational studies and small trials have established that positive fluid balance, volume overload, and elevated filling pressures in patients with sepsis, AKI, or other critical illnesses are associated with poor outcomes including progressive kidney dysfunction, impaired AKI recovery, increased risk of subsequent sepsis, impaired wound healing, prolonged mechanical ventilation, and death.^{8,10,218-255} These data also suggest that fluid overload may dilute serum creatinine and thereby mask the diagnosis of AKI.^{95,256} These studies have led many to postulate that fluid overload directly impairs kidneys and other organ function,²⁵⁷⁻²⁵⁹ but whether fluid overload causes poor outcomes or this association is confounded by

severity of illness (i.e., sicker patients simply require larger volumes of fluid) remains unclear.

In this context, the concept of fluid restriction or deresuscitation has gained recognition in the critical care field. However, apart from possibly the setting of ARDS,⁹⁴ prospective data demonstrating the benefit of fluid restriction in the ICU are limited. Despite promising results of pilot studies and smaller trials,^{220,242,260-262} thus far large RCTs of fluid restriction in sepsis, major surgery, or other critical illness have not shown benefit. The 2018 Restrictive versus Liberal Fluid Therapy in Major Abdominal Surgery (RELIEF) trial randomized 3000 patients undergoing major abdominal surgery to two different fluid regimens and demonstrated that fluid restriction resulted in no benefit with increased rates of AKI and need for KRT.²⁶³ More recently, two large multicenter RCTs, the Conservative versus Liberal Approach to Fluid Therapy in Septic Shock (CLASSIC) and the Crystalloid Liberal or Vasopressors Early Resuscitation in Sepsis (CLOVERS) trials, analyzed the effects of fluid restriction in septic shock and demonstrated that fluid restriction was safe but failed to improve outcomes.^{264,265} Though additional large multicenter RCTs are ongoing (e.g., Australasian Resuscitation In Sepsis Evaluation: FLUID or vasopressors In emergency Department Sepsis [ARISE FLUIDS, NCT04569942] and Early Vasopressors in Sepsis [EVIS, NCT05179499] trials), high-quality evidence to guide the use of IV fluids in the ICU remains limited.

Nonetheless, guidelines based largely on observational data continue to emphasize the need for *early* resuscitation in septic shock.^{248,250,266-276} This has led to the conceptualization of four temporal phases of fluid therapy in sepsis or critical illness (Fig. 64.1).²⁷⁷⁻²⁷⁹ Consistent with this concept are trial data showing that fluid responsiveness disappears quickly, typically within hours, in patients with septic shock treated with modern resuscitation practices.²⁸⁰

Nephrologists are often tasked with helping achieve deresuscitation in patients with AKI requiring KRT. However, no large prospective studies exist to guide the optimal approach to the deresuscitation phase.²⁸¹⁻²⁸³ Observational data are conflicting, with some studies suggesting that the ability to achieve negative fluid balance with CRRT is associated with higher survival, primarily in patients with some degree of fluid overload at the time of CRRT initiation, and others finding that higher CRRT fluid removal rates are associated with increased mortality.^{233,284-289} Importantly, the gap between prescribed and achieved fluid removal with CRRT is an underrecognized parameter associated with increased mortality in patients with AKI and fluid overload.²⁸⁴ This lack of data has contributed to substantial worldwide practice variation in fluid management with KRT.²⁹⁰⁻²⁹³ Though pilot studies specifically analyzing approaches to ultrafiltration with CRRT are ongoing (NCT05306964, NCT05473143) and the use of point-of-care ultrasound and other novel technologies to guide fluid removal hold promise, additional data are clearly needed.^{294,295}

CHOICE OF INTRAVENOUS FLUIDS

CHOICE OF INTRAVENOUS FLUIDS: SALINE VERSUS BALANCED CRYSTALLOIDS

Observational²⁹⁶⁻³⁰⁰ and animal data^{301,302} suggest that 0.9% saline, when compared with balanced salt solutions such

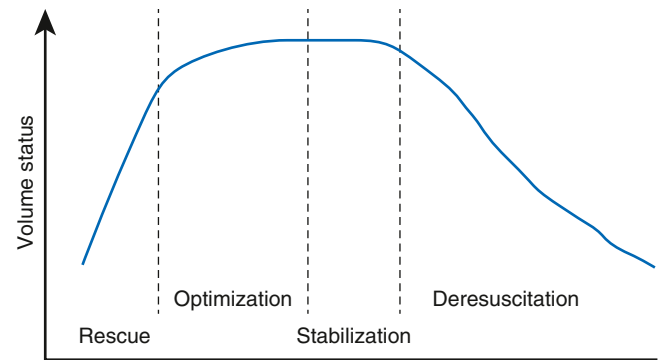


Fig. 64.1 Four phases of fluid therapy in sepsis or other critical illness: “ROSE.” Optimal fluid resuscitation strategies should parallel the phases of septic shock or other critical illnesses. The first phase (e.g., first 6 hours) is **Resuscitation**, when initial aggressive resuscitation, recommended to be 30 mL/kg ideal body weight of crystalloid in most patients,²⁷⁶ should be provided. The second phase is **Optimization**, when additional fluids are given specifically targeting fluid responsiveness (i.e., increased stroke volume or cardiac output) or other specific hemodynamic endpoints to improve organ function. The third phase is **Stabilization**, also referred to as the “tone and time” phase, in which patients remain on vasopressors but vasopressor requirements are no longer increasing, patients are no longer fluid responsive, organ function is stable or improving, and further fluids are withheld. The fourth phase is **Evacuation**, also known as the deresuscitation or mobilization phase, when patients with volume overload are treated with diuretics or ultrafiltration to expedite the return to euvolemia in hopes of accelerating recovery from critical illness.²⁷⁷⁻²⁷⁹ The time at which a given patient transitions between phases varies, and repeated insults may substantially disrupt this sequence. (Reproduced with permission from ADQI (www.ADQI.org).)

as lactated Ringer (LR) or Plasma-Lyte (PL, Baxter), may increase the risk of AKI, need for KRT, and mortality. The pathophysiology underlying the nephrotoxicity of saline is thought to be related to the supraphysiologic chloride content and the resultant hyperchloremic acidosis that saline induces. Proposed mechanisms include reduced glomerular filtration rate via activation of the tubuloglomerular feedback triggered by increased chloride delivery to the macula densa, vasoconstriction caused by chloride-induced thromboxane release, and increased inflammatory cytokine expression induced by acidosis.³⁰¹⁻³⁰³ Imaging studies on human volunteers demonstrate that saline infusion reduces RBF.³⁰⁴

Since 2010, a series of large clinical trials have analyzed crystalloid choice in the ICU, starting with a 2012 before-and-after study and a 2015 cluster crossover RCT, which yielded conflicting results.^{305,306} In 2018, two large, single-center, pragmatic, unblinded, multiple-crossover cluster trials were published analyzing the use of LR or PL vs. saline, one in patients admitted from the ED to a non-ICU bed, the Saline against Lactated Ringer or Plasma-Lyte in the Emergency Department (SALT-ED) trial³⁰⁷ ($n > 13,000$), and another in ICU patients, the Isotonic Solutions and Major Adverse Renal Events Trial (SMART; $n > 15,000$).³⁰⁸ Both studies demonstrated that balanced solutions resulted in an approximate 1% reduction in the composite endpoint of major adverse kidney events (a composite of death, need for KRT, or persistent doubling of creatinine above baseline) by 30 days or discharge.

These two single-center studies were followed in 2021 and 2022 by two large multicenter RCTs. First, the Balanced Solutions in Intensive Care Study (BaSICS) in Brazil randomized >11,000 individual patients in 75 ICUs to the blinded use of PL or saline whenever possible throughout the entire ICU course and found no difference in 90-day mortality.³⁰⁹ There was also no difference in subgroups with sepsis or AKI, but the risk of death was significantly increased with LR in the subgroup with traumatic brain injury (TBI). Though methodologically rigorous, BaSICS was criticized for being underpowered despite the large sample size due to lower-than-predicted mortality, relatively low acuity of the subjects, administration of nonstudy fluids before-and-after randomization, and relatively low total volumes of fluids administered, which resulted in statistically significant but modest differences in serum chloride levels (e.g., 2–3 mEq/L higher in the saline arm at 48 hours).^{309,310} The subsequent Plasma-Lyte 148 versus Saline (PLUS) study, carried out in 53 ICUs in Australia and New Zealand, was similar in design, with the major exceptions being that the study excluded TBI and was aborted after enrollment of 5037 of the planned 8800 patients due to the COVID-19 pandemic.³¹¹ Otherwise, though PLUS also had some contamination by nonstudy fluid, the severity of acute illness and volumes of study fluid administered were somewhat higher than in the SMART or BaSICS trials. Like BaSICS, the primary outcome of 90-day mortality was not significantly different in the PL and saline arms. Soon after the publication of PLUS, a Bayesian meta-analysis of 13 RCTs including >35,000 ICU patients (of which >30,000 come from SMART, BaSICS, and PLUS) was published and concluded that the risk ratio for 90-day mortality with balanced crystalloids versus saline was 0.96 (95% confidence interval [CI], 0.91–1.01; $I^2=12.1\%$), with similar effects on risk of AKI and need for KRT.³¹² Large multicenter RCTs of PL versus saline in patients undergoing cadaveric renal transplantation ($n=808$) and in pediatric septic shock ($n=708$) have demonstrated benefits in kidney function with PL.^{313,314}

Though somewhat more expensive, there are no data (outside of TBI) to suggest harm from balanced solutions, and, as such, the use of balanced solutions seems reasonable in most ICU settings. Alternatively, one can reasonably interpret the BaSICS and PLUS trials as evidence that saline is largely safe in the ICU. Importantly, saline will continue to be the fluid of choice in patients with TBI or hypovolemic hypochloremic metabolic alkalosis.

CHOICE OF INTRAVENOUS FLUIDS: COLLOIDS VERSUS CRYSTALLOIDS

There are no compelling data to support the routine use of colloids over crystalloids as primary resuscitation fluid in ICU patients given a series of negative large RCTs^{315–319} and the increased expense of colloids. Two of these, the Saline versus Albumin Fluid Evaluation (SAFE)³¹⁵ and Albumin Italian Outcome Sepsis (ALBIOS)³¹⁹ trials, have shown albumin to be a safe alternative to crystalloids in ICU patients. Both trials suggested a possible benefit of lower mortality with albumin in subgroups with sepsis, via a nonsignificant trend in SAFE and a significant post hoc subgroup analysis in ALBIOS.^{315,319} Likewise, most^{320–322} (but not all³²³) meta-analyses have concluded that albumin use is associated with reduced mortality in sepsis. Additional RCTs evaluating the

role of albumin in the treatment of sepsis, such as albumin as endothelial rescue therapy, are ongoing (NCT03869385) or planned³²⁴ including the ALBIOS-BALANCED trial (NCT03654001).

On the basis of a statistically significant increase in mortality rate in the subgroup with TBI in SAFE,³¹⁵ albumin use should be avoided in TBI patients, for which saline is considered the fluid of choice. The use of artificial colloids, such as hydroxyethyl starch, gelatins, and dextrans, should be avoided in ICU patients due to an increased risk of AKI and a lack of evidence for their use.^{316–318,325,326}

USE OF INTRAVENOUS BICARBONATE IN THE INTENSIVE CARE UNIT

Metabolic acidosis is common in the ICU, present in 6% to 42% of patients, and, when persistent, is associated with a mortality of >50%.^{327,328} Using IV sodium bicarbonate to correct metabolic acidosis in the ICU, though commonly done, remains controversial.^{328–332} Despite the theoretical benefits of improved hemodynamic responsiveness to catechol vasopressors, animal models,^{333,334} human observational data,^{335,336} and small human trials^{337–339} suggest little to no benefit by attempting to correct acidemia with supplemental IV bicarbonate, with a possible signal in observational data for improved outcomes in patients with sepsis complicated by AKI.³³⁶ Potential harms of exogenous sodium bicarbonate administration include intracellular acidosis, cerebral spinal fluid acidosis, impaired oxygen offloading in tissues, stimulation of lactic acidosis, disinhibition of glycolysis, hypercarbia and/or depression of respiratory drive, hypokalemia, hypocalcemia, overshoot alkalosis, volume overload, and (when given in hypertonic solutions) hyponatremia.^{340,341}

The highest-quality data addressing the use of IV bicarbonate in the ICU available thus far come from the Sodium Bicarbonate to Treat Severe Acidosis in the Critically Ill (BICAR-ICU) trial.³⁴² The study randomized nearly 400 patients in 26 ICUs in France with severe metabolic or mixed acidosis ($\text{pH} \leq 7.20$, arterial $\text{pCO}_2 \leq 45$ mm Hg, serum bicarbonate ≤ 20 mmol/L) to unblinded treatment with 4.2% (500 mmol/L) sodium bicarbonate or standard care. Patients with pure respiratory acidosis, ketoacidosis, prior alkali therapy or KRT, stage IV CKD, and proven digestive or urinary tract loss of sodium bicarbonate were excluded, with the latter two criteria functionally excluding most non-anion gap metabolic acidosis. The primary outcome was the composite of death at 28 days or ≥ 1 organ failure at 7 days and was not statistically significant. However, patients in the subgroup with stage 2 or 3 AKI ($n=182$) had significant decreases in the primary outcome and both of its components, with a 28-day mortality of 46% vs. 63% ($P=0.017$) favoring sodium bicarbonate administration. In addition, there was a reduction in the need for KRT in the sodium bicarbonate group (35% vs. 52%). Though this subgroup analysis was prespecified and adjusted for multiple comparisons, the results remain hypothesis-generating. The findings of BICAR-ICU support the cautious use of sodium bicarbonate in critically ill patients with severe acidosis and AKI and the general avoidance of liberal sodium bicarbonate use in ICU patients without AKI.^{330–332} Larger studies are ongoing to better inform the use of sodium bicarbonate,

specifically in ICU patients with AKI including the BICAR-ICU-2 trial³⁴³ (NCT04010630) and the U.K. Multicentre Evaluation of Sodium Bicarbonate in Acute Kidney Injury in Critical Care (MOSAICC) trial (ISRCTN14027629).

KIDNEY REPLACEMENT THERAPY

The care of patients experiencing AKI is primarily supportive, with KRT required in about 5% to 10% of critically ill individuals with severe AKI.^{344,345} The goals of KRT in AKI include maintaining fluid balance, providing metabolic control, and preventing uremia-related complications, while simultaneously minimizing potential harm to the kidneys and facilitating renal recovery. KRT options include IHD, CRRT, hybrid therapies like prolonged intermittent renal replacement therapy (PIRRT), and peritoneal dialysis (PD). Despite these interventions, severe AKI requiring KRT is associated with a 50% to 60% mortality rate, prolonged hospital stay, and development of CKD and end-stage kidney disease.^{4,346-352}

TIMING OF KIDNEY REPLACEMENT THERAPY

Determining when to initiate KRT has been a subject of debate.³⁵³⁻³⁵⁵ In cases of severe AKI with medically refractory complications, initiating KRT is recommended to alleviate the deterioration of nonkidney organ function and reduce the risk of death. These urgent indications include severe hyperkalemia, severe metabolic acidosis, fluid overload resistant to diuretic therapy, signs of uremia, and specific drug intoxications.³⁵⁶⁻³⁵⁸ The specific criteria and thresholds for metabolic acidosis, hyperkalemia, or pulmonary edema vary widely in clinical practice.

Over the years, small, underpowered RCTs aimed to determine the optimal timing for initiating KRT, yielding mixed results.³⁵⁹⁻³⁶⁵ Since 2016, five well-designed RCTs addressed timing (Table 64.2). The trial Early versus Delayed Initiation of Renal Replacement Therapy on Mortality in Critically Ill Patients with Acute Kidney Injury (ELAIN) randomized 231 KDIGO stage 2 AKI patients to immediate or delayed KRT based on reaching KDIGO stage 3 AKI or developing life-threatening indications. The early KRT group had significantly lower 90-day mortality (39.3% vs. 54.7%) with a hazard ratio (HR) of 0.66 in favor of early KRT.³⁶⁶ In contrast, the Artificial Kidney Initiation in Kidney Injury (AKIKI),³⁶⁷ Initiation of Dialysis Early versus Delayed in the Intensive Care Unit (IDEAL-ICU),³⁶⁸ and Standard versus Accelerated Initiation of Renal Replacement Therapy in AKI (STARRT-AKI)³⁶⁹ trials failed to show an advantage to early initiation. The AKIKI trial enrolled 620 KDIGO stage 3 AKI patients, with no survival difference between early and delayed KRT.³⁶⁷ IDEAL-ICU, which included 448 AKI patients with septic shock, was terminated early due to no significant mortality difference.³⁶⁸ In the STARRT-AKI study, 2927 patients with KDIGO stage 2 or 3 AKI were randomized into an accelerated KRT group (initiation within 12 hours) or a standard group, where KRT was discouraged until AKI had persisted for more than 72 hours or an urgent indication developed (eTable 64.1).³⁶⁹ The study showed no 90-day mortality difference but more KRT-dependency and adverse events with early KRT.

The AKIKI2 study investigated whether adopting a strategy of delayed intervention, as initially defined in the first AKIKI trial, had inferior outcomes compared with an even more delayed approach. In this trial, 278 patients with oliguria for >72 hours or a BUN >112 mg/dL were randomized to immediate KRT initiation or a more delayed strategy, requiring a mandatory indication or a BUN >140 mg/dL.³⁷⁰ The primary outcome, days alive and free of KRT at day 28, did not differ significantly. However, multivariable analysis revealed a higher HR for death at 60 days with the more delayed strategy (1.65; 95% CI 1.09–2.50, $P = 0.018$).

The combined results from these trials indicate that, without specific indications, initiating KRT early does not offer clinical advantages and might impede renal recovery. Nonetheless, the AKIKI2 trial suggests there may be a threshold at which delaying KRT might result in adverse effects. The Acute Disease Quality Initiative workgroup on CRRT recommends a more personalized approach for acute KRT that does not rely solely on kidney function or AKI stage, but it also considers the patient's clinical condition (eTable 64.2).³⁵⁶ The workgroup proposes starting KRT when metabolic and fluid demands exceed kidney capacity, influenced by the severity of acute illness, solute load, fluid accumulation, and underlying chronic diseases.³⁵⁶ However, the specific components of the demand/capacity concept require further definition and validation.

KIDNEY REPLACEMENT THERAPY MODALITY

KRT modalities remove water and solutes by transport across a semipermeable membrane. Ultrafiltration, responsible for plasma water removal, operates through hydrostatic pressure (as in IHD, PIRRT, and CRRT) or an osmotic gradient (as in PD). Solute removal during KRT employs diffusion and/or convection. In extracorporeal treatments, certain solutes may also be eliminated by adsorption onto the membrane.

Convection, as occurs during ultrafiltration, involves the movement of solutes by bulk flow (Fig. 64.2A). When plasma water passes through the semipermeable membrane, it effectively “drags” small- and middle-molecular-weight solutes. Pore size and membrane charge determine the size of the solutes removed, and solute clearance is generally proportional to the ultrafiltration rate. In contrast, diffusion drives solutes across the membrane via concentration gradients between blood and dialysate (Fig. 64.2B). Running the dialysate countercurrent to blood flow maximizes this gradient. Smaller solutes like urea diffuse more effectively than larger ones such as β_2 -microglobulin, as solute diffusivity is inversely proportional to molecular weight. All KRT modalities utilize ultrafiltration for fluid removal and diffusion, convection, or both for solute clearance.

INTERMITTENT HEMODIALYSIS

IHD sessions are typically prescribed three times weekly, lasting 3 to 4 hours each, with blood flow rates of 300 to 500 mL/min and dialysate flow rates of 500 to 800 mL/min (Table 64.3). Ultrafiltration removes excess fluid, and solute clearance mainly relies on diffusion, with some larger solutes cleared via convection due to increased membrane porosity and internal filtration and backfiltration.³⁷¹ The frequency and duration of dialysis depend on patient metabolic control, volume status, and hemodynamic stability. Initial

Table 64.2 Randomized Controlled Trials Evaluating Timing of Kidney Replacement Therapy Initiation for Critically Ill Adult Patients With Acute Kidney Injury

	ELAIN (2016)³⁶⁶	AKIKI (2016)³⁶⁷	IDEAL-ICU (2018)³⁶⁸	STARRT-AKI (2020)³⁶⁹	AKIKI-2 (2021)³⁷⁰
Setting	Single center in Germany	31 Centers in France	29 Centers in France	168 Centers in 15 countries	39 Centers in France
N	231	620	488	2927	278
Inclusion criteria	KDIGO stage 2 AKI	KDIGO stage 3 AKI	RIFLE FAILURE stage AKI	KDIGO stage 2 or 3 AKI	KDIGO stage 3 with oliguria >72 h or BUN 112-140 mg/dL
Timing of earlier KRT	Within 8 h	Within 6 h	Within 12 h	Within 12 h	Within 12 h
Timing of later KRT	Within 12 h of KDIGO stage 3 AKI or “absolute indications” ^a	Life-threatening complications of AKI ^b , BUN >112 mg/dL, or oliguria >72 h	48 h post-randomization in the absence of kidney recovery	Life-threatening complications of AKI ^c or AKI persisting >72 h	Life-threatening complications of AKI ^b or BUN >140 mg/dL
Median time to KRT (earlier vs. later)	6 h vs. 25.5 h	2 h vs. 57 h	7.6 h vs. 51.5 h	6.1 h vs. 31.1 h	3 h vs. 33 h
% who received KRT (earlier vs. later)	100% vs. 91%	98% vs. 51%	97% vs. 62%	97% s. 62%	98% vs. 79%
Primary outcome (earlier vs. later)	90-day mortality: 39.3% vs. 54.7% (<i>P</i> = 0.03)	60-day mortality: 48.5% vs. 49.7% (<i>P</i> = 0.79)	90-day mortality: 58% vs. 54% (<i>P</i> = 0.38)	90-day mortality: 43.9% vs. 43.7% (<i>P</i> = 0.92)	KRT-free days within 28 days: 12 vs. 10 days (<i>P</i> = 0.93)
Other key outcomes	Shorter KRT duration and hospital length of stay in early arm	Spontaneous diuresis occurred earlier in delayed arm	No difference in ICU or hospital length of stay	KRT dependence at 90 days higher in early arm (10% vs. 6%, 95% CI 1.2-2.4)	In prespecified adjusted analysis, more delayed strategy was associated with mortality (<i>P</i> = 0.02)
KRT-related complications	No difference	More catheter-related blood stream infections in early arm	More hyperkalemia in delayed arm	More adverse events ^d (17% vs. 23%, <i>P</i> < 0.001) in early arm	No difference
Limitations	Limited generalizability (smaller sample, single center, mostly surgical)	55% of patients were treated initially with IHD	Stopped early due to futility	Decision for KRT initiation ultimately left to provider discretion	Small sample size

^aIndications for KRT initiation in the delayed arm in ELAIN included BUN >100 mg/dL, potassium >6 mEq/L or hyperkalemia-induced ECG changes, magnesium >9.7 mg/dL, anuria or oliguria with urine output <200 mL per 12 h, or organ edema resistant to ≥1 dose of loop diuretic.

^bIndications for KRT initiation in the delayed arm in AKIKI and AKIKI-2 included potassium >6 mEq/L, potassium >5.5 mEq/L despite medical therapy, metabolic acidosis with pH <7.15, and pulmonary edema resulting in hypoxemia despite diuretic therapy.

^cIndications for KRT initiation in the delayed arm in STARRT-AKI included potassium ≥6.0 mEq/L, pH ≤7.20 or serum bicarbonate ≤12 mEq/L, and volume overload with hypoxemia.

^dIn STARRT-AKI, the early arm experienced statistically significant increases in hypotension and hypophosphatemia, trends for more bloodstream infections and arrhythmias (*p* = 0.07 for both), but no significant difference in serious adverse events (*P* = 0.15).

BUN, Blood urea nitrogen; CI, confidence interval; IHD, intermittent hemodialysis; KDIGO, Kidney Disease: Improving Global Outcomes;

sessions typically use lower blood and dialysate flows for patients susceptible to dialysis disequilibrium.³⁷²

IHD is preferred for rapid correction of electrolyte imbalances and drug intoxications due to its rapid solute removal. However, IHD can lead to intradialytic hypotension in 10% to 70% of acute treatments.³⁷³⁻³⁷⁷

Proposed interventions for improving hemodynamic stability are mostly extrapolated from patients on maintenance IHD. These interventions include decreasing the ultrafiltration rate, lowering the blood flow rate, sodium modeling, cooling the dialysate, increasing the dialysate calcium concentration, using isolated ultrafiltration, using

eTable 64.1 Summary of Standard versus Accelerated Initiation of Renal Replacement Therapy in Acute Kidney Injury (STARRT-AKI) Trial³⁶⁹**Setting:**

- Multicenter: 168 hospitals in 15 countries in 5 continents
- October 2015 to September 2019

Key inclusion criteria:

- ICU admission
- Severe AKI (KDIGO stage 2 or 3)
- Need for clinical equipoise (i.e., the investigator felt that neither initiation nor deferral of KRT was mandated)^a

Study design and study arms:

- Open-label RCT with 1:1 allocation
- Accelerated strategy (KRT initiated within 12 hours of meeting eligibility): $n = 1462$
- Standard care (KRT discouraged unless specific criteria met): $n = 1465$

Patient characteristics at randomization:

- Mean age 65
- 32% female
- 44% with underlying chronic kidney disease
- 67% medical ICU admission
- 58% with sepsis
- 77% requiring mechanical ventilation
- 70% requiring vasoactive support
- Mean serum creatinine 3.5 mg/dL

Initial KRT modality (among KRT-treated patients)

- 69% CRRT
- 26% IHD
- 4.4% SLED

Primary outcome (accelerated arm vs. standard arm):

- No difference in 90-day mortality by modified intention-to-treat analysis (43.7% vs. 43.9%, RR 1.0, $P = 0.92$)

Key secondary outcomes (all reported as accelerated arm vs. standard arm):

- 96.8% vs. 61.8% initiated KRT
- Higher rate of KRT dependence at 90 days of 10.4% vs. 6.0% (RR 1.74, 95% CI 1.24-2.43)
- Higher rate of adverse events (AEs) of 23.0% vs. 16.5% ($P < 0.001$), with difference driven by the 2 most common AEs, hypotension and hypophosphatemia
- Higher rate of readmission at 90 days of 20.9% vs. 17.0% (RR 1.23, 95% CI 1.02-1.49)
- No difference in serious AEs (1.0% vs. 0.5%, $P = 0.15$)
- No difference in primary outcome in 6 prespecified subgroups: gender, baseline eGFR, severity of illness (i.e., SAPS II score), presence or absence of sepsis, surgical or medical ICU admission, and geographic region

^aOf >23,000 patients screened that otherwise met inclusion criteria, >7,800 were excluded on basis of lack of equipoise.

CI, Confidence interval; CRRT, continuous renal replacement therapy; eGFR, estimated glomerular filtration rate; ICU, intensive care unit; IHD, intermittent hemodialysis; KDIGO, Kidney Disease: Improving Global Outcomes; KRT, kidney replacement therapy; RCT, randomized controlled trial; RR, relative risk; SAPS II, Simplified Acute Physiology II score; SLED, sustained low-efficiency dialysis.

eTable 64.2 Factors to Consider When Deliberating Timing of Kidney Replacement Therapy (KRT) Initiation for Acute Kidney Injury (AKI) Patients in the Intensive Care Unit (ICU)

Illness Severity and Trajectory

- AKI severity and trend
- Fluid balance and symptoms of fluid overload
- Severity of electrolyte and acid-base disorders
- Presence of relevant nonrenal organ dysfunction impacted by AKI and/or fluid overload
- Anticipated need for enhanced metabolic support (e.g., rhabdomyolysis or tumor lysis syndrome)

Necessity of KRT

- Presence of oliguria, considering the effect of diuretics
- Likelihood of recovery of kidney function without KRT, considering the reversibility of the specific etiology of AKI, trends in kidney function and urine output, and baseline kidney function
- Underlying comorbidities impacted by AKI and/or fluid overload

Risks of KRT

- Hemodynamic instability from KRT
- Infection, including catheter-related bloodstream infection
- Other risks associated with vascular access (pneumothorax, procedural bleeding, catheter-associated deep venous thrombosis, etc.)
- Clearance of trace elements, water-soluble vitamins, phosphate, amino acids/small peptides, and drugs (especially antibiotics)
- Delayed renal recovery
- Increased risk of immobilization / impeding physical rehabilitation

Other Factors

- Patient and family wishes and overall goals of care
- Overall prognosis including likelihood of patient survival
- Availability of machines, disposable supplies, and nursing staff
- Health care costs

Adapted from Macedo E, Mehta RL. Continuous dialysis therapies: core curriculum 2016. *Am J Kidney Dis.* 2016;68(4):645–657 and Ostermann M, Joannidis M, Pani A, et al. Patient selection and timing of continuous renal replacement therapy. *Blood Purif.* 2016;42(3):224–237.

hypertonic infusions, extending the duration of dialysis or increasing the frequency of treatments in order to slow the ultrafiltration rate, and using midodrine.³⁷⁸⁻³⁹⁰ In general, these interventions lack robust evidence, especially in critically ill patients with AKI. Additionally, the rapid shifts of intracellular fluid and solutes during IHD can result in increased ICP and cerebral edema in patients with acute brain injury.³⁹¹

CONTINUOUS RENAL REPLACEMENT THERAPY

CRRT offers an alternative for managing hemodynamically unstable patients with AKI intolerant to IHD. CRRT uses diffusion, convection, or a combination to remove solutes and fluids gradually. This method theoretically allows for better hemodynamic tolerance compared with IHD by providing slower solute and fluid removal per unit time. CRRT is generally performed 24 hours a day with blood flow rates typically between 100 and 300 mL/min (Table 64.3). Advantages of CRRT include better hemodynamic tolerance, continuous solute clearance, and improved volume control (Fig. 64.3).²²³ Drawbacks include vascular access challenges, filter clotting, increased anticoagulation needs, reduced patient mobility, and higher costs and ICU nurse workload compared with IHD. Although rare, dialysis disequilibrium has been reported with CRRT.^{392,393}

CRRT modalities for solute removal are continuous venovenous hemofiltration (CVVH), continuous venovenous hemodialysis (CVVHD), and continuous venovenous hemodiafiltration (CVVHDF). CVVH relies on convection without dialysate, using a hemofiltration solution (often referred to as “replacement fluid” based on historical practices) infused before the filter (prefilter) or after the filter (postfilter) to restore volume and electrolytes. When using postfilter hemofiltration solution, the filtration fraction should be kept <0.25 to reduce the likelihood of clotting of the hemofilter.³⁹⁴ This can be done by increasing the blood flow rate³⁹⁵ or introducing some hemofiltration solution before the filter, although the latter will dilute the concentration of solutes entering the hemofilter, leading to a decrease in solute clearance.^{396,397} CVVHD primarily uses diffusion with slower dialysate flow rates, resulting in near-complete saturation of the dialysate. Ultrafiltration is primarily used for volume control. CVVHDF combines convection and diffusion, utilizing both dialysate and replacement fluid.

Despite enhanced middle-molecule clearance in convective therapies, meta-analyses comparing hemofiltration to hemodialysis for AKI found no significant differences in patient survival.³⁹⁸ Although high-volume hemofiltration (HVHF) with ultrafiltration rates >50 mL/kg/h has been used to enhance cytokine clearance in AKI patients with sepsis, controlled studies have not consistently demonstrated better outcomes.^{30,364,399,400} With no clear advantage of one CRRT modality over another, choice should be based on clinician preference and expertise, with attention to circuit survival.

PROLONGED INTERMITTENT RENAL REPLACEMENT THERAPY

Prolonged intermittent renal replacement therapy (PIRRT), an alternative to CRRT for critically ill patients

with AKI, offers a hybrid approach.⁴⁰¹⁻⁴⁰³ These techniques mitigate the hemodynamic impact of IHD by extending the duration of therapy and adapting conventional hemodialysis machines to operate at lower blood pump speeds (100–300 mL/min) and dialysate flow rates (100–300 mL/min) (Table 64.3). Alternatively, CRRT equipment can be adapted for PIRRT, with sessions lasting 6 to 18 hours. PIRRT practices vary substantially in technology, prescription, and anticoagulation.^{403,404} Diffusion-based PIRRT is described by terms like “sustained low-efficiency (daily) dialysis” (SLED or SLEDD), “extended daily dialysis” (EDD), “slow continuous dialysis” (SCD), and “go slow dialysis.” When combining convection and diffusion, these therapies are known as low-efficiency daily diafiltration (SLEDD-f) and sustained hemodiafiltration (S-HDF). “Accelerated venovenous hemofiltration” is used for exclusively convective therapy.⁴⁰⁴⁻⁴⁰⁷

PIRRT combines advantages of CRRT and IHD, offering improved hemodynamic stability by gradually removing solutes and fluids over an extended timeframe. It eliminates the need for 24-hour therapy and costly CRRT equipment and solutions. PIRRT provides high solute clearances similar to IHD, allowing for uninterrupted diagnostic and therapeutic procedures.^{402,404} It can also be administered overnight, promoting patient mobility during the day. Studies confirmed hemodynamic control comparable with CRRT.^{404,408-411} In some medical institutions, PIRRT is employed to transition patients from CRRT to conventional IHD when their hemodynamic stability improves.^{354,404}

PERITONEAL DIALYSIS

PD leverages the peritoneum as a semipermeable membrane for solute removal by diffusion. A catheter is used to instill a dialysate solution into the peritoneal cavity, where it dwells for a specified duration, allowing solutes to diffuse from the blood into the dialysate before drainage. The dialysate contains varying high-glucose concentrations to establish an osmotic gradient for fluid removal. Acute PD can be conducted intermittently or continuously and either manually or through an automated cycler. PD techniques for AKI encompass high-volume PD, acute intermittent peritoneal dialysis (AIPD), continuous equilibrated peritoneal dialysis (CEPD), tidal peritoneal dialysis (TPD), and continuous flow peritoneal dialysis (CFPD).

High-volume PD employs an automated cycler to deliver 18 to 22 exchanges of 2 L every 24 hours, offering the highest small-solute clearance and ultrafiltration rate among PD methods.^{412,413} AIPD employs brief dwell times with dialysate flows of 1 to 4 L/hour for 12 to 24 hours every other day. CEPD resembles continuous ambulatory PD for chronic outpatient PD but without patient ambulation.^{414,415} In TPD, a portion of dialysate remains in the peritoneal cavity, facilitating rapid exchanges through a cycler, and improving solute clearance.⁴¹⁶ CFPD involves simultaneous inflow and outflow of dialysate through two routes, maintaining a fixed intraperitoneal volume, with high dialysate flow rates and urea clearances of 30 to 50 mL/min.^{417,418}

PD offers advantages like technical simplicity, hemodynamic stability, reduced dialysis disequilibrium risk, no need for anticoagulation or vascular access, and cost-effectiveness.^{419,420} Gradual solute and fluid

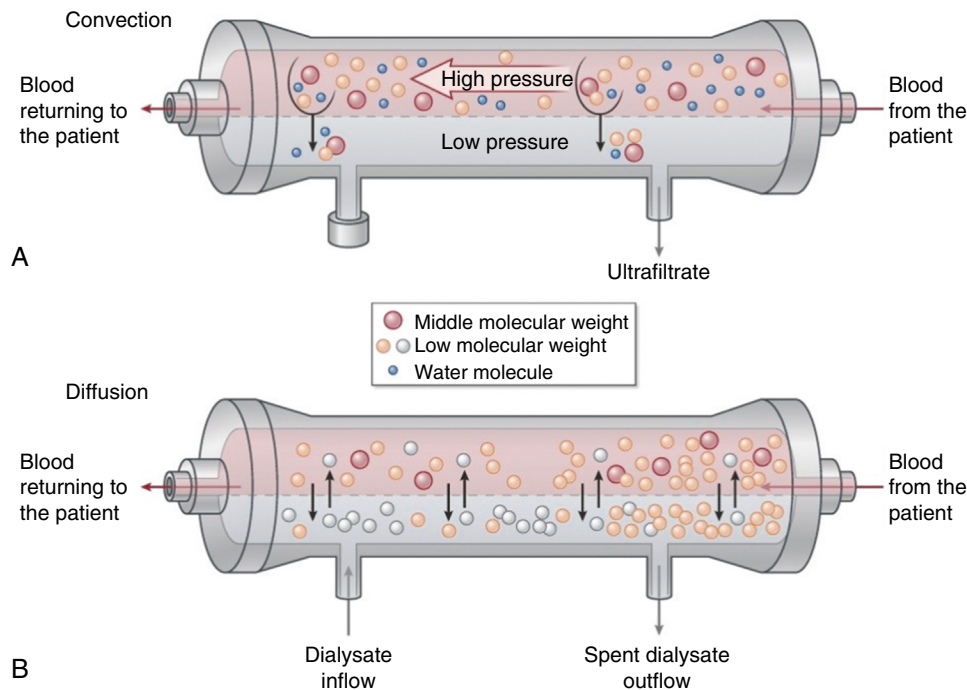


Fig. 64.2 Modalities of kidney replacement therapy use primarily convective clearance (hemofiltration), diffusive clearance (hemodialysis), or both. (A) In hemofiltration, solute clearance occurs primarily by convection. In convection, solutes are transported across the hemofilter membrane along with plasma water as a result of hydrostatic pressure (i.e., transmembrane pressure) generated on the blood side of the membrane. Solute cleared by convection include urea and other small molecules along with larger “middle molecules.” (B) In hemodialysis (HD), solute clearance occurs primarily by diffusion, which is driven by a concentration gradient across the semipermeable membrane. Small solutes in high concentration in the blood diffuse across the membrane into the dialysate, which contains either little (e.g., potassium) or none (e.g., urea) of the solutes being cleared. Small solutes in higher concentration in the dialysate (e.g., bicarbonate) diffuse into the blood. Dialysate runs across the HD membrane countercurrent to the direction of blood flow to maintain a concentration gradient for removal of small solutes along the entire length of the semipermeable membrane. Modern hemodialyzers are virtually all “high-flux” dialyzers, which clear substances larger than historical low-flux dialyzers. However, unlike hemofiltration, HD does not effectively clear larger middle molecules. Ultrafiltration can be performed with HD by applying a transmembrane pressure across the membrane, but, in contrast to the high volume of ultrafiltration used to achieve significant solute clearance in hemofiltration, the volumes of ultrafiltration performed in HD are relatively small, contribute little to solute clearance, and are instead used only to achieve net volume removal. KRT, Kidney replacement therapy. (Reproduced with permission from Teixeira JP, Neyra JA, Tolwani A. Continuous KRT: a contemporary review. *Clin J Am Soc Nephrol.* 2023;18[2]:256–269.)

removal makes it safer for patients at risk of intracranial hypertension. Drawbacks include specialized catheter training, catheter placement complications, peritonitis risk, potential solute inadequacy, unpredictable ultrafiltration, hyperglycemia, protein loss, and respiratory compromise due to intraabdominal pressure.^{419,420} PD is contraindicated for individuals with recent abdominal surgery, abdominal drains, diaphragmatic disease, or ileus.^{419,420}

SELECTION OF KIDNEY REPLACEMENT THERAPY MODALITY

While CRRT is often considered standard of care for hemodynamically unstable patients needing KRT, no specific KRT modality has been proven to be superior in AKI.⁴²¹ Existing data from RCTs^{422–427} and meta-analyses^{428–434} generally regard CRRT and IHD as complementary therapies, with comparable outcomes including mortality and kidney recovery.^{15,435} Some trials suggest improved hemodynamic tolerance with CRRT, but others report no significant differences in hypotension rates or vasoactive support.^{422,424,427} PIRRT also exhibits comparable outcomes with CRRT regarding mortality, renal recovery, ICU length of stay, and fluid removal.^{409–411,436–438} However, phosphate

removal has been noted to be higher in PIRRT compared with CRRT.⁴³⁹

Small RCTs support PD as an effective AKI treatment with outcomes akin to IHD and CRRT in terms of efficacy and survival.^{440–442} A Cochrane review involving 484 patients with AKI requiring dialysis found no significant differences in mortality, recovery of kidney function, or infectious complications between PD and other KRT modalities.⁴⁴² In a subsequent trial, 125 critically ill patients with AKI were randomly assigned to high-volume tidal PD or CRRT. At 28 days, PD-treated patients had higher survival rates (70% vs. 47%, $P < 0.01$) and renal recovery rates (60% vs. 37%, $P < 0.01$), along with fewer infectious complications and shorter ICU stays.⁴⁴³ Overall, acute PD is valuable in patients with hemodynamic instability and vascular access challenges or in places with limited KRT resources.

In clinical practice, hemodynamically stable patients often receive IHD, while hemodynamically unstable, fluid-overloaded, or septic patients receive CRRT or PIRRT.^{15,444–446} CRRT might improve volume management more effectively than conventional IHD²²³ yet appears comparable with PIRRT.⁴³⁷ CRRT preserves cerebral perfusion, making it preferable for patients at risk of cerebral edema.^{447,448} Additionally, CRRT

Table 64.3 Comparison of Kidney Replacement Therapy Modalities

	Intermittent Hemodialysis (IHD)	Prolonged Intermittent Renal Replacement Therapy (PIRRT)^a	Continuous Renal Replacement Therapy (CRRT)
Typical parameters	- Duration: 3–4 h - Frequency: 3–4 times per week Q_b: 300–400 mL/min Q_d: N/A Q_r: 600–800 mL/min UF_{NET} rate: 0–10 mL/kg/h	- Duration: 6–12 h - Frequency: 3–6 times per week Q_b: 200–300 mL/min Q_d: N/A Q_r: 200–300 mL/min UF_{NET} rate: 0–8 mL/kg/h	- Duration/frequency: continuous Q_b: 100–300 mL/min Q_r: none or 16–50 mL/min Q_d: none or 16–50 mL/min UF_{NET} rate: 0–2 mL/kg/h
Advantages	- Widespread availability - Low cost of disposables - Delivery with no anticoagulation usually feasible - Most rapid instantaneous clearance - May be performed with catheter or arteriovenous fistula or graft - Lowest rate of hypophosphatemia - Time off KRT allows more time for physical therapy relative to CRRT	- Low cost of disposables - Delivery with no anticoagulation usually feasible - Likely improved hemodynamic tolerance and fluid removal relative to IHD - May be performed with catheter or arteriovenous fistula or graft - Time off KRT allows more time for physical therapy relative to CRRT	- Likely best hemodynamic stability - Most effective for fluid management - Least likely to exacerbate intracranial hypertension - Superior solute control (i.e., higher total daily or weekly dose) - Usually permits nutrition without restriction in protein, phosphate, or potassium - Facilitates gradual controlled correction of dysnatremia - Possibly increased renal recovery among survivors^b
Disadvantages	- Likely highest risk of hypotension - Likely highest risk of exacerbating intracranial hypertension - Limited fluid removal - Often requires limitation in dietary potassium or phosphate intake	- Least standardization of prescription - Lack of large randomized controlled trials validating optimal prescription or comparing efficacy to IHD or CRRT - Limited data on drug dosing, especially antimicrobials - Limited data on safety in acute brain injury	- High cost of disposables/highest overall cost - Lowest instantaneous solute clearance - Greatest need for anticoagulation - Highest risk of hypophosphatemia - Requires catheter placement^c - Increased risk of immobilization^d

^aPIRRT encompasses a variety of hybrid modalities such as sustained low-efficiency dialysis (SLED), accelerated venovenous hemofiltration (AVVH), and many others; note that the typical PIRRT parameters shown are for SLED and that the listed advantages and disadvantages may not apply equally to all types of PIRRT.

^bThe increased rate of renal recovery noted in survivors of AKI initially treated with CRRT relative to IHD is based on observational data only and therefore controversial.

^cAlthough one single-center observational study⁵¹⁷ suggests that using arteriovenous fistulae/grfts for CRRT may be safe and feasible, doing so is not standard of care in most institutions.

^dCRRT is not a contraindication to early mobilization per se as observational data⁵¹⁸ suggest that physical rehabilitation is feasible and safe in patients on CRRT, but many providers perceive CRRT to be a barrier to physical therapy.

Q_b , blood flow rate; Q_d , dialysate flow rate; Q_r , replacement fluid rate; UF_{NET} , net ultrafiltration rate.

offers the unique advantage of gradual and precise correction of dysnatremias.^{449,450} It is also suitable for conditions requiring ongoing solute control (e.g., severe rhabdomyolysis and tumor lysis syndrome). In contrast, IHD is preferred for rapid solute removal, as in severe hyperkalemia or drug intoxications.³⁹⁵ PIRRT and IHD allow intermittent treatment breaks, facilitating physical therapy and early mobilization. Ultimately, the selection of KRT modality should consider available resources, cost, provider expertise, and patient needs.

PRESCRIBING KIDNEY REPLACEMENT THERAPY

VASCULAR ACCESS

According to the 2012 KDIGO AKI guidelines, the preferred site for temporary dialysis catheter placement is the right internal jugular (IJ) vein, followed by the femoral vein,

followed by the left IJ, with the subclavian veins as a last resort.¹⁵ The catheter should be inserted with the tip in a large vein at an appropriate depth: 15 to 20 cm for right IJ catheters and 20 to 24 cm for left IJ catheters with a tip at the cavoatrial junction and at least 20 to 25 cm for femoral catheters with a tip in the inferior vena cava.^{451,452} Ultrasound guidance and strict aseptic technique are recommended to minimize complications.¹⁵ While there is consensus favoring the right IJ, some argue for left IJ over femoral placement, especially in obese patients at higher risk of femoral catheter complications.⁴⁵³ In an RCT comparing femoral and jugular catheters, infection rates were similar overall but higher in the femoral group for patients with a BMI >28.4 kg/m².⁴⁵³ In a post hoc analysis, right IJ placement had the lowest catheter malfunction rate, followed by femoral, with highest in the left IJ, though the malfunction of the left IJ catheters in this study may have been due to inadequate depth of

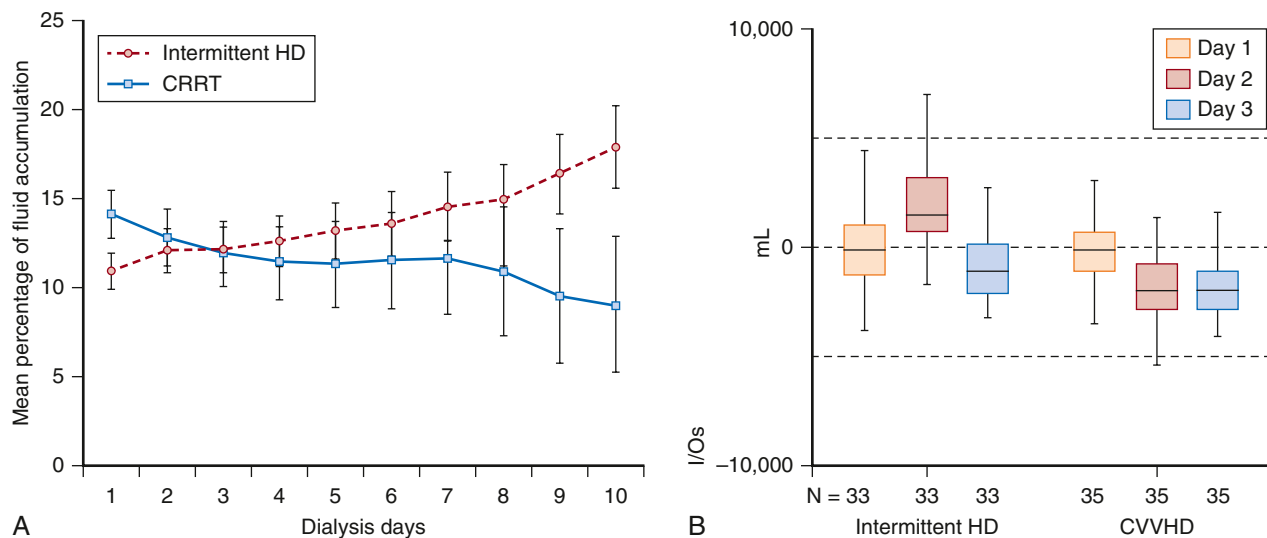


Fig. 64.3 Superior volume control with CRRT versus intermittent hemodialysis (IHD). (A) Fluid accumulation over time in patients on CRRT and on IHD. Data from the PICARD (Program to Improve Care in Acute Renal Disease) study. (B) Twenty-four-hour total fluid balance (I/Os; milliliters) during the first 3 days of kidney replacement therapy in patients on IHD and CVVHD therapy. Shown are median values with interquartile range (box borders) and extreme values (whiskers). CRRT, Continuous renal replacement therapy; CVVHD, continuous venovenous hemodialysis; IHD, intermittent hemodialysis. (A reproduced with permission from Bouchard J, Soroko SB, Chertow GM, et al. Fluid accumulation, survival and recovery of kidney function in critically ill patients with acute kidney injury. *Kidney Int.* 2009;76[4]:422–427. B reproduced with permission from Augustine JJ, Sandy D, Seifert TH, Paganini EP. A randomized controlled trial comparing intermittent with continuous dialysis in patients with ARF. *Am J Kidney Dis.* 2004;44[6]:1000–1007.)

insertion (16 cm).^{451,454} Observational studies suggest lower infection and thrombosis risks with tunneled catheters, along with better blood flow rates.^{377,455} Therefore considering a tunneled dialysis catheter over a nontunneled one should be considered for prolonged KRT requirements, but further research is needed before making it a standard practice for AKI in the ICU.^{377,456}

DOSE

KRT dosing in AKI is determined by Kt/V_{urea} and frequency in IHD and PIRRT, as well as effluent flow rate (mL/kg/h) in CRRT. Two large RCTs, the VA/NIH ATN study and the RENAL study, evaluated KRT dosing in AKI. In the ATN trial, intensive KRT included high-dose CVVHDF or PIRRT for unstable patients and IHD for stable patients. In the intensive approach, patients received CVVHDF with an effluent rate of 35 mL/kg/h or six PIRRT/IHD sessions per week, with each session targeting a Kt/V_{urea} of 1.2 to 1.4.⁴⁵⁷ The less intensive approach used CVVHDF with an effluent rate of 20 mL/kg/h or three PIRRT/IHD treatments per week with the same Kt/V_{urea} target. Notably, the intensive KRT strategy did not reduce 60-day mortality or improve kidney recovery. The RENAL trial compared CRRT at doses of 40 and 25 mL/kg/h and revealed no survival advantage with higher CRRT intensity at 90 days.⁴⁵⁸ Multiple meta-analyses also showed no survival benefit with increased KRT intensity.^{459–461} Studies of high-volume CRRT with doses up to 70 to 85 mL/kg/h have similarly shown no benefit.^{30,364}

On the basis of these studies, KDIGO 2012 AKI guidelines recommend a CRRT-delivered dose of 20 to 25 mL/kg/h and a Kt/V_{urea} of ≥ 1.2 per session for PIRRT or IHD.^{15,446} In IHD, continued patient assessment is needed to ascertain whether additional treatments are required to maintain electrolyte and acid-base balance and control volume status.⁴⁴⁶ For CRRT, doses of 25 to 30 mL/kg/h

can be considered to account for therapy interruptions.¹⁵ Temporary use of higher doses may be required for specific conditions including hyperkalemia, severe acidosis, or hyperammonemia, but dose should be adjusted back to standard ranges once solute control is achieved.³⁹⁵ CRRT dosing should remain adaptable and continuously adjusted to meet the dynamic needs of critically ill patients.

High-intensity KRT can have adverse consequences by removing beneficial solutes, such as essential nutrients, antiinflammatory cytokines, and antibiotics.^{15,462} Both the ATN and RENAL studies observed increased rates of hypophosphatemia with higher KRT intensity.^{457,458} In the ATN study, higher-intensity KRT was associated with lower rates of successful extubation and fewer ventilator-free days, especially in patients with lower serum phosphate levels.⁴⁶³ Higher KRT intensity also increased the risk of a $\geq 50\%$ decline in urine output (HR = 1.29; 95% CI, 1.1–1.51).⁴⁶⁴ Patients treated only with IHD in the ATN study had more KRT-free days with lower-intensity KRT.⁴⁶⁵ A meta-analysis of six trials involving 1926 subjects found that higher-intensity KRT was associated with longer KRT duration and a higher likelihood of KRT dependence at day 28 (HR = 1.15, 95% CI 1–1.33).⁴⁶¹ Additionally, high-intensity KRT is costlier due to expenses for disposable items and nursing labor.^{462,466} The threshold for harmful underdosing in KRT intensity is uncertain. Some experts suggest a potential mortality risk below 20 mL/kg/h, but observational data from Japan suggest that doses as low as 15 mL/kg/h are safe.⁴⁶⁷

ANTICOAGULATION

KDIGO 2012 guidelines for AKI recommend using regional citrate anticoagulation (RCA) over heparin in CRRT unless contraindicated.¹⁵ Citrate chelates ionized calcium (iCa) in the extracorporeal circuit, and optimal anticoagulation is achieved when hemofilter iCa is <0.4 mmol/L.⁴⁶⁸ A portion

of the calcium-citrate complex is lost across the hemofilter while the rest enters the systemic circulation, where the liver metabolizes citrate to bicarbonate and calcium is released into the circulation.⁴⁶⁹ Calcium is infused into the patient to replace the calcium lost across the hemofilter.

Proactive management of RCA is essential to mitigate potential side effects including metabolic alkalosis and acidosis, hypocalcemia and hypercalcemia, and hypomagnesemia.⁴⁶⁹⁻⁴⁷² Patients with severe shock liver and lactic acidosis may not effectively metabolize citrate.^{469,473,474} Citrate accumulation is characterized by low systemic iCa, elevated total calcium, total calcium to systemic iCa ratio >2.5, anion gap metabolic acidosis, and escalating calcium infusion requirements (Table 64.4).^{472,475-477} Nonetheless, RCA has been safely used in advanced liver disease and perioperative liver transplant patients.⁴⁷⁸⁻⁴⁸⁰

Multiple RCTs including the multicenter Effect of Regional Citrate Anticoagulation versus Systemic Heparin Anticoagulation During Continuous KRT on Dialysis Filter Life Span and Mortality Among Critically Ill Patients With Acute Kidney Injury (RICH) trial support RCA's superiority over systemic heparin in extending filter life and reducing bleeding risk.⁴⁸¹⁻⁴⁸³ In the RICH trial, exploratory secondary outcomes demonstrated a potential association between citrate and an increased risk of infection.^{482,483} However, a post hoc analysis found that filter lifespan exceeding 48 h, but not anticoagulation choice, was associated with higher infection rates.⁴⁸⁴ Further research is needed to clarify this observation.

Anticoagulation-free KRT sessions are feasible with IHD and PIRRT due to the higher blood flow and shorter durations. CRRT without anticoagulation may require more circuit changes due to clotting, but strategies such as higher blood flow rates and prefilter replacement solution may

mitigate clotting risk.⁴⁸⁵⁻⁴⁸⁸ In one RCT, different blood flow rates (150 vs. 250 mL/min) in CRRT had no significant impact on filter clotting, possibly due to higher flow rates triggering more pressure alarms and reductions or stoppages in blood pump speed, with consequent increased clotting risk.⁴⁸⁹

ULTRAFILTRATION

Achieving euvolemia is crucial in critical illness, especially due to the harms of fluid overload.^{223,254,490} Ultrafiltration and rapid osmotic shifts during IHD can cause intradialytic hypotension from inadequate plasma refill and impaired hemodynamic compensatory mechanisms.³⁷⁷ Additionally, IHD can induce hypotension due to "myocardial stunning," unrelated to ultrafiltration, observed in both patients on maintenance IHD and those with AKI undergoing IHD.^{491,492} Surprisingly, myocardial stunning was also seen in one study of patients with AKI on CRRT, suggesting microcirculatory factors might play a role.⁴⁹³

Data on prescribing net ultrafiltration (UF_{NET}) in CRRT are limited. UF_{NET} is the net volume of fluid removed per hour adjusted for patient body weight (mL/kg/h).⁴⁹⁴ In a secondary analysis of the RENAL trial, UF_{NET} rates above 1.75 mL/kg/h were associated with lower survival and a higher risk of hypophosphatemia.²⁸⁵ Accumulating observational data suggest a nonlinear relationship between UF_{NET} and mortality, with the lowest mortality at UF_{NET} rates of 1.0 to 1.75 mL/kg/h (Fig. 64.4).^{285,494} Iterative assessment of patient's fluid balance (desired vs. achieved) and tolerance to fluid removal is key for risk stratification.^{284,495} Prospective studies are needed to validate the proactive targeting of UF_{NET} within the 1.0 to 1.75 mL/kg/h range and/or other methods to achieve personalized fluid balance goals.³⁹⁵

Table 64.4 Metabolic Complications of Regional Citrate Anticoagulation Related to Excess or Inadequate Citrate

Complication	Mechanism	Diagnostic Features	Management Options
Excess buffer from citrate	Excess citrate supply, resulting in excessive metabolic conversion of citrate to bicarbonate and excess buffer	- Metabolic alkalosis $tCa/iCa < 2.5$	- Decrease blood flow rate/citrate rate^a - Increase dialysate rate - Decrease buffer concentration in other CRRT solutions
Citrate accumulation ^b	Impaired metabolic conversion of citrate resulting in accumulation of citrate-calcium complexes in blood	- Anion gap metabolic acidosis $tCa/iCa > 2.5$ - Escalating calcium infusion requirements	- Decrease blood flow rate/citrate rate^a - Increase dialysate rate - Discontinue regional citrate anticoagulation
Insufficient buffer from citrate	Inadequate citrate supply, resulting in insufficient conversion of citrate to bicarbonate and insufficient buffer	- Metabolic acidosis $tCa/iCa < 2.5$	- Increase blood flow rate/citrate rate^a - Decrease dialysate rate - Increase buffer concentration in other CRRT solutions

^aThese changes in blood flow and citrate rate are done proportionally, to increase or decrease the delivery of citrate while maintaining the same ratio of citrate infusion rate to blood flow rate and thus similar intrafilter ionized calcium concentration.

^bCitrate accumulation is also known as citrate toxicity or citrate lock.

CRRT, Continuous renal replacement therapy; iCa, ionized calcium; tCa, total calcium.

Adapted with permission from Neyra JA, Yessayan L, Thompson Bastin ML, et al. How to prescribe and troubleshoot continuous renal replacement therapy: a case-based review. *Kidney360*. 2021;2(2):371–384.

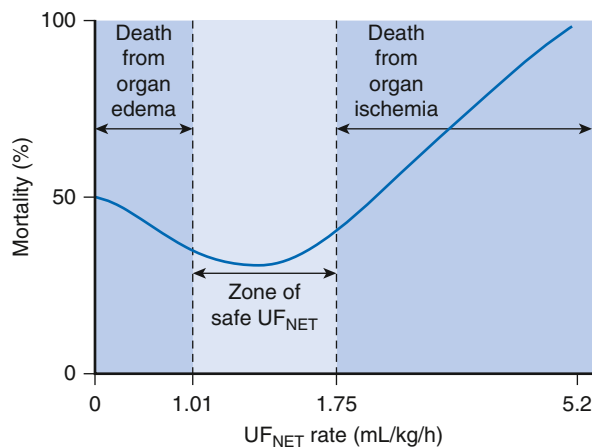


Fig. 64.4 Net ultrafiltration rate is independently associated with mortality in patients treated with continuous renal replacement therapy (CRRT) in a nonlinear fashion. The relationship between mortality and net ultrafiltration (UF_{NET}) in patients in the intensive care unit with acute kidney injury (AKI) is J shaped. In a post hoc analysis of >1400 patients from the Randomized Evaluation of Normal versus Augmented Level Replacement Therapy (RENAL) study comparing high versus low dose of CRRT for AKI in the intensive care unit,⁴⁵⁸ low UF_{NET} rates <1.01 mL/kg per hour, and high UF_{NET} rates >1.75 mL/kg per hour were associated with higher risk-adjusted 90-day mortality compared with UF_{NET} rates in a middle range of 1.01–1.75 mL/kg per hour. It has been proposed that these effects are mediated by harms of organ edema in those treated with low UF_{NET} rates and by organ ischemia in those treated with high UF_{NET} rates, but no prospective trial data yet exist to demonstrate that targeting a moderate rate of UF_{NET} improves outcomes. (Data from Murugan R, Kerti SJ, Chang CH, et al. Association of net ultrafiltration rate with mortality among critically ill adults with acute kidney injury receiving continuous venovenous hemodiafiltration: a secondary analysis of the randomized evaluation of normal vs augmented level [RENAL] of renal replacement therapy trial. *JAMA Netw Open*. 2019;2[6]:e195418. Reprinted with permission from Murugan R, Bellomo R, Palevsky PM, Kellum JA. Ultrafiltration in critically ill patients treated with kidney replacement therapy. *Nat Rev Nephrol*. 2021;17[4]:262–276.)

DRUG-DOSING CONSIDERATIONS

Specific drug-dosing considerations in critically ill patients with AKI include attention to drugs with long half-lives and those that rapidly achieve therapeutic concentrations. Importantly, loading doses do not typically need to be adjusted in patients with AKI or those requiring KRT. In contrast, maintenance doses require adjustments in patients with AKI or those requiring KRT. When KRT is initiated, dosing strategies must further account for the extracorporeal clearance of medications. Drug characteristics (e.g., volume of distribution, molecular weight, and extent of protein binding), KRT prescriptions (e.g., blood flow, session duration, proportion of convective and diffusive clearances), and dialyzer features (e.g., porosity or flux, surface area or efficiency) are important determinants of medication clearance. In some cases, KRT may have a more subtle impact on drug clearance by enhancing extrarenal drug metabolism, perhaps through the removal of uremic toxins.⁴⁹⁶ For recipients of IHD, medications should be administered after dialysis or a postdialysis supplemental dose should be considered. With the predominant use of high-flux filters, which enhance drug clearance, dosing

recommendations that were derived for use with low-flux filters typically need to be increased by 25% to 50%.⁴⁹⁶

The advent of PIRRT poses a further challenge because extracorporeal clearance is augmented by this modality as compared with IHD. This may result in increased drug removal and inadequate blood concentrations of vital medications including antibiotics. To date, drug-dosing guidance with PIRRT is available for only a few agents.⁴⁹⁷ As a general rule, drug doses should probably be higher than those administered to patients receiving IHD and, where feasible, their use should be accompanied by careful drug level monitoring.⁴⁹⁸ A “top-up” dose following the conclusion of a PIRRT session is advisable for agents that are likely to be dialyzable.

In patients receiving CRRT, total effluent dose and the relative breakdown of convective and diffusive therapy will affect drug removal and subsequent dosing changes. CRRT at usual doses, assuming no interruptions to therapy, begins to approximate endogenous kidney function (~30 to 60 mL/min), and more frequent drug dosing is generally required than with IHD or PIRRT. For a more detailed discussion of this topic, the reader is referred to [Chapter 56](#).

EXTRACORPOREAL BLOOD PURIFICATION

Extracorporeal blood purification (EBP) entails removal of pathogens, endotoxins, and inflammatory mediators from the blood and has evolving applications in critically ill patients with multiorgan failure and sepsis.⁴⁹⁹ Among the various EBP technologies, adsorptive and immunomodulatory EBP devices have gained interest.

In the setting of infection, immune cells recognize pathogen-associated membrane proteins (PAMPs) that activate the production of antiinflammatory and proinflammatory cytokines. Injured host cells also express damage-associated molecular patterns, which cause leukocyte activation and cytokine production.^{499,500} EBP technologies using adsorption provide means to modulate a dysregulated immune system by removal of inflammatory mediators and highly protein-bound toxins that are not effectively cleared by diffusion or convection. Adsorption consists of removal of molecules via attachment to various sorbents given chemical affinity to ion exchange resins and chemisorbents.^{501,502}

Traditionally, adsorptive-based hemoperfusion has been used for the treatment of intoxications (e.g., paraquat poisoning) and extracorporeal liver support devices.⁵⁰³ Liver support devices, such as the molecular adsorbent recirculating system, remove albumin-bound solutes and have been used for refractory cholestatic pruritus or drug intoxications, especially when accompanied by severe liver injury and for acute or acute-on-chronic liver failure as a bridge to liver transplantation.⁵⁰³⁻⁵⁰⁶

The use of novel adsorptive-based EBP devices was accelerated during the COVID-19 pandemic. These new technologies include filters for direct pathogen removal (Seraph 100 Microbind, GARNET, Hemopurifier),^{507,508} endotoxin removal (Toraymyxin),⁴⁹⁹ cytokine removal (CytoSorb, oXiris, SepXiris),^{499,509} and immunomodulatory devices such as the selective cytopheretic device (SCD).⁵¹⁰ SCD is a synthetic membrane cartridge that binds and

deactivates activated leukocytes and therefore mitigates cytokine production.

While scientific progress in EBP is promising, outstanding questions related to clinical utility, patient selection, timing of initiation, duration of treatment, heterogeneity of treatment effect, and others remain unanswered.^{13,511} Importantly, the application of adsorptive-based nonselective clearance to remove both proinflammatory and antiinflammatory mediators may not be beneficial if applied at the wrong time, given that some patients may develop immune dysfunction or immunoparalysis characterized by impaired cytokine production and subsequent increased risk of infection.^{10,11,512,513} Nonetheless, extensive research is underway to address key questions about how various extracorporeal therapies and the interactions among the host, pathogen/insult, and medical devices affect patient-centered outcomes in critical illness and multiorgan failure.⁵¹⁴

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QUESTIONS

This vignette applies to questions 1 and 2.

A 54-year-old man with diabetes and hypertension is admitted to the hospital with left lower lobe pneumonia. Despite appropriate antibacterial therapy, the following morning he is transferred to the ICU with shock and worsening respiratory distress requiring mechanical ventilation. Vital signs include temperature 38.7°C; BP 102/58 mm Hg on norepinephrine at 0.1 µg/kg/min; and HR 92 beats per min. Serum creatinine is 1.8 mg/dL (baseline 0.9 mg/dL), and lactate is 5.2 mmol/L. Arterial blood gas reveals pH 7.28, pCO₂ 42 mm Hg, and PaO₂ 128 mm Hg on 100% oxygen with positive end-expiratory pressure of 10 cm H₂O. A chest radiograph shows progression of his lung disease with interval development of patchy bilateral opacities, and he is diagnosed with acute respiratory distress syndrome. Lung protective ventilation with low tidal volumes (6 mL/kg ideal body weight) is initiated.

1. What is the most common complication of this ventilation strategy?
 - a. Increased risk of barotrauma leading to pneumothorax
 - b. Hypoxemia leading to increased risk of acute kidney injury and need for kidney replacement therapy
 - c. Respiratory acidosis due to permissive hypercapnia
 - d. Increased intrathoracic pressures resulting in impaired venous return and hypotension requiring additional IV fluids
 - e. Increased risk of ventilator dyssynchrony resulting in increased need for neuromuscular blockade

Answer: c

Rationale: ARDS is frequently associated with impaired ventilation (i.e., increased ventilatory dead space and impaired CO₂ excretion). When lung protective ventilation is used for ARDS, respiratory acidosis due to permissive hypercapnia frequently results. Especially in the setting of AKI, patients may develop severe acidemia from combined metabolic and respiratory acidosis for which supplemental bicarbonate or KRT may be indicated. Lung protective ventilation targets lower plateau airway pressures and tidal volumes and will tend to *decrease* the risk of barotrauma. As shown in the original ARDS Net trial⁸⁷, lung protective ventilation does result in somewhat more hypoxemia, but lung protective ventilation nonetheless results in decreased mortality, underlying the fact that the primary goal of mechanical ventilation for ARDS is not to completely normalize arterial oxygen or CO₂ but rather to minimize the risk of ventilator-induced lung injury while maintaining adequate (but not necessarily normal) gas exchange. Furthermore, in the ARDS Net trial, no signals for increased risk of organ dysfunction were seen in the lung protective ventilation arm, with a trend toward *improved* kidney function, consistent with animal and epidemiologic data, which suggest that injurious mechanical ventilation may increase the risk of AKI.⁸⁸⁻⁹⁰ While the positive intrathoracic pressures generated by all mechanical ventilation may decrease venous return, clinically significant hemodynamic impairment is relatively uncommon. Furthermore, lung protective ventilation does not necessarily decrease venous return more than

other ventilatory strategies, with the effect on venous return depending on the level of PEEP applied and the mean airway pressures achieved. Ventilator dyssynchrony can be a complication of the lower tidal volumes used with lung protective ventilation, but if it occurs, adjustment to the ventilator settings or increased sedation should be tried before implementation of neuromuscular blockade. Though pharmacologic paralysis is necessary in some patients with ARDS, a large multicenter RCT⁵²⁰ demonstrated no mortality benefit to protocolized neuromuscular blockade in ARDS when compared with standard light sedation used in modern practice.

2. Despite appropriate treatment and lung protective ventilation, the patient's hypoxemia worsens and, ultimately, he is initiated on venovenous extracorporeal membrane oxygenation (VV-ECMO) on ICU Day 3. With ECMO, his gas exchange improves, allowing for implementation of "rest" settings on the ventilator. In addition, his shock resolves, allowing him to come off vasopressor therapy; his lactate normalizes; and his AKI improves, with serum creatinine returning to his baseline of 0.9 mg/dL. However, on ECMO Day 5, he develops a leukocytosis, with white blood cell count of 26,000/mm³, and, though he remains off of vasopressors, develops recurrent AKI. Blood cultures are drawn and bronchoscopy with bronchoalveolar lavage (BAL) is performed. On ECMO Day 8 blood and BAL cultures both grow *Pseudomonas aeruginosa*. However, despite appropriate antimicrobial coverage, his AKI worsens, with serum creatinine increasing to 3.1 mg/dL. Despite high-dose IV furosemide, his urine output drops to 346 mL per 24 h, resulting in a net positive fluid balance of >2 L per day. Potassium is 5.6 mEq/L and plasma-free hemoglobin is 34 mg/dL (upper limit of normal 30 mg/dL). Which of the following is the next best step in management?
 - a. Hold off on KRT initiation as the subgroup of ECMO patients in the STARRT-AKI (Standard versus Accelerated Initiation of Renal Replacement Therapy in Acute Kidney Injury) trial had no benefit to early KRT.
 - b. Initiate intermittent hemodialysis (IHD) because it has been shown in trials to be superior to CRRT for management of fluid balance in ICU patients with AKI.
 - c. Obtain renal ultrasound, urinalysis, and fractional excretion of sodium (FENa) because sepsis is unlikely to cause AKI in the absence of hypotension requiring vasopressors.
 - d. Recommend ECMO circuit change as plasma-free hemoglobin above the upper limit of normal implicates excess hemolysis as the cause of AKI.
 - e. Initiate standard-dose continuous KRT (CRRT) for volume management as positive fluid balance is associated with poor outcomes in patients on ECMO.

Answer: e

Rationale: Though experts⁶⁶ recommend against the *preemptive* use of KRT in patients supported with ECMO, fluid overload in these patients is independently associated with an increased risk of mortality⁵⁹⁻⁶⁴ and timely KRT should be

considered in ECMO patients with AKI because they are unlikely to tolerate prolonged oliguria in the setting of cardiopulmonary failure. In this case, given the failure to respond to high-dose loop diuretic resulting in positive fluid balance and some degree of hyperkalemia, initiation of KRT at this time appears reasonable. RCT data to guide the initiation of KRT in specifically ECMO patients are lacking because patients requiring ECMO were not included in STARTRT-AKI or the other recent RCTs analyzing the timing of KRT initiation in the ICU. There are no compelling data to suggest that CRRT results in superior long-term outcomes, such as survival or renal recovery, relative to IHD when used in patients in the ICU with AKI. However, CRRT has been unequivocally shown to be superior in treating or preventing fluid accumulation, as the continuous modality affords 24 hours per day (rather than 4 hours every 1–2 days) to carry out gradual ultrafiltration. As such, where both modalities are available, the preferential use of CRRT over IHD in patients requiring ECMO is reasonable. With the possible exception of FENa, the diagnostic tests recommended in answer c are reasonable. However, sepsis can cause AKI without hypotension, especially in the setting of unequivocal sepsis, such as this scenario with bacteremia with a compelling pathogen, prominent leukocytosis, and new organ dysfunction. While ischemic and septic AKI often coexist in critically ill humans, pathophysiologic studies demonstrate that septic and ischemic AKI are distinct entities as macrovascular renal blood flow may be normal or *high* early in the course of septic AKI. In general, FENa performs poorly in differentiating prerenal azotemia from ATN in the ICU,¹⁰⁶ and, as pigment nephropathy can cause ATN characteristically with a low FENa, it may be especially misleading in this case where some hemolysis is present. However, while the plasma-free hemoglobin is high, it is only marginally elevated in this case. The plasma-free hemoglobin level at which circuit change is necessary is poorly defined by the literature, and the decision to perform an ECMO circuit exchange is complex, but typically much higher elevations in plasma-free hemoglobin (e.g., several times the upper limit of normal) are necessary to justify the potential risk and expense of a circuit change.

3. A 38-year-old woman with cirrhosis is admitted for hepatic encephalopathy (HE) triggered by spontaneous bacterial peritonitis (SBP). She is admitted to the ward and treated with ceftriaxone, IV albumin, and lactulose, with initial improvement in her mental status. On Hospital Day 4 she develops recurrent deterioration in mental status, tachypnea, and AKI with serum creatinine rising within 48 hours to 2.1 mg/dL (above a baseline of 0.6) for which nephrology is consulted and IV albumin is restarted. Vital signs include BP 84/54 with mean arterial pressure (MAP) 64 mm Hg and oxygen saturation of 92% of 4 L/min of oxygen. A chest radiograph shows a new right middle lobe opacification for which her antibiotics are broadened to treat nosocomial pneumonia. Examination reveals jaundice, grade 3 HE, moderately tense abdomen with obvious ascites, and 3 to 4+ generalized pitting edema. Laboratory values include total bilirubin 6.8 mg/dL, serum albumin 3.2 g/dL, INR 2.6, and lactate 1.8 mmol/L. Urine studies include FENa of <0.1%, bland urinalysis, and undetectable urine protein

and urine albumin. Renal ultrasound is normal, and she has had no nephrotoxin exposure. What is the next most appropriate step to treat her AKI?

- Stop IV albumin and transfer the patient to the ICU for initiation of norepinephrine targeted to a MAP of 80 to 85 mm Hg for suspected hepatorenal syndrome–AKI (HRS-AKI).
- To avoid ICU transfer, she should be placed on continuous pulse oximetry and telemetry monitoring and started on terlipressin on the ward.
- Given the FENa <1%, she likely has prerenal azotemia and therefore should be continued on IV albumin at a dose of 20 to 40 gm IV daily.
- Given that she has a new pneumonia, she does not meet diagnostic criteria for HRS-AKI and therefore is not a candidate for vasoconstrictor therapy, but she should be transferred to the ICU as she is likely to have cerebral edema.
- Given her AKI, therapeutic paracentesis is contraindicated regardless of the severity of her ascites, but since HRS-AKI is the most common cause of AKI in patients with liver disease, she should be started on octreotide and midodrine for her HRS-AKI if she fails to improve with 48 hours of albumin 1 mg/kg IV daily.

Answer: a

Rationale: Though HRS-AKI is overall less common than ATN or prerenal azotemia as a cause of AKI in the setting of cirrhosis,¹⁰³ this patient is likely to have HRS-AKI. She meets nearly all the formal criteria for HRS-AKI^{99,105} including AKI by KDIGO criteria, cirrhosis with ascites, absence of shock or nephrotoxin exposure, and unremarkable urine studies and renal ultrasound. Additional clinical features that strongly suggest HRS-AKI include low MAP and very low FENa (with <0.2% or less being suggestive of HRS-AKI and *not* prerenal azotemia). Though the absence of active infection or sepsis was previously required to diagnose HRS, the newer criteria only preclude a diagnosis of HRS-AKI in the setting of septic shock, as it is now better recognized that infection may trigger HRS-AKI that persists despite appropriate treatment of the infection. Traditional criteria also require failure to respond to 48 hours of IV albumin therapy. However, while a trial of IV albumin is appropriate in most patients with AKI and cirrhosis, assessment of volume status before starting albumin is prudent and albumin should not be given to patients like this one with overt volume overload or significant hypoxemia.^{100–102,122} In addition, albumin does not treat HRS-AKI. Given this presentation is overall highly suggestive of HRS-AKI, delaying vasoconstrictor therapy for 48 hours to evaluate response to albumin may be harmful as response to vasoconstrictors appears best when initiated early.^{99,108,113} She has two contraindications to terlipressin therapy including hypoxemia of any severity and multiorgan dysfunction, with AKI with serum creatinine ≥ 2 mg/dL, grade 3 HE, and coagulopathy with INR ≥ 2.5 (i.e., acute-on-chronic liver failure [ACLF] grade ≥ 3). As such, given that octreotide and midodrine are generally ineffective for HRS, the best option is transfer to the ICU for norepinephrine therapy, which, though not studied in as many total patients, has been shown in multiple studies to have similar efficacy for HRS-AKI as terlipressin.^{99,108,113} Cerebral edema is a potential lethal complication of ALF, being present in a substantial minority

of patients with ALF and grade 3 HE and the majority of ALF patients with grade 4 HE.¹²³ Cerebral edema is characteristically absent in cirrhosis.^{126,127} With new multiorgan dysfunction in the setting of cirrhosis, this patient meets criteria for ACLF, which may occasionally be complicated by cerebral edema, but it is not characteristic of ACLF. Finally, the role of intraabdominal hypertension in HRS-AKI is being increasingly appreciated and, though some caution may be prudent (e.g., limitation of volume removed to ≤ 5 L), careful therapeutic paracentesis is not contraindicated in the setting of kidney dysfunction and may have some potential to *improve* renal function via decreased intraabdominal pressure and increased renal perfusion pressure.^{118-121,139}

4. A 62-year-old man with AKI secondary to septic shock has been initiated on CRRT. He is mechanically ventilated and requires two vasopressors for blood pressure support. His weight is 80 kg and CRRT dose is 25 mL/kg/h. His urine output is 20 mL/h. Within 12 hours, his serum creatinine has decreased from 4.5 mg/dL to 3.8 mg/dL, potassium has decreased from 6.5 mEq/L to 5.2 mEq/L, and bicarbonate has increased from 16 mEq/L

to 22 mEq/L. What is the appropriate next step in the management of this patient?

- a. Decrease the CRRT dose to 20 mL/kg/h.
- b. Increase the CRRT dose to 35 mL/kg/h.
- c. Maintain the current CRRT dose of 25 mL/kg/h.
- d. Discontinue CRRT.

Answer: c

Rationale: The appropriate next step in this patient's management is to maintain the current CRRT dose of 25 mL/kg/h. The recommended minimum dose of CRRT is 20 to 25 mL/kg/h. However, to account for treatment interruptions and the natural decline in filter efficiency over time, guidelines suggest prescribing 25 to 30 mL/kg/h of effluent dose.^{15,395,471} The patient's serum creatinine has decreased since initiation of CRRT, and he has improving serum potassium and bicarbonate levels, indicating that the current dose is effectively removing solutes from the blood. Increasing the CRRT dose may lead to excessive removal of solutes and has not been shown to improve outcomes. Decreasing the CRRT dose may not be sufficient to adequately remove solutes. Discontinuing CRRT at this time may not be appropriate, given the ongoing renal injury, oliguria, and need for continued support.